Autodesk®
Scaleform®

Scaleform CLIK AS3

Scaleform CLIK

Prasad Silva, Matthew Doyle

2.0

2010 8 19

Autodesk®
GAMEWARE

Copyright Notice

Autodesk® Scaleform® 4.3

© 2013 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD, AutoCAD Learning Assistance, AutoCAD LT, AutoCAD Simulator, AutoCAD SQL Extension, AutoCAD SQL Interface, Autodesk, Autodesk 123D, Autodesk Homestyler, Autodesk Intent, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, Beast (design/logo), BIM 360, Built with ObjectARX (design/logo), Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Cleaner Central, ClearScale, Colour Warper, Combustion, Communication Specification, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, Design Doctor, Designer's Toolkit, DesignKids, DesignProf, Design Server, DesignStudio, Design Web Format, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, Exposure, Extending the Design Team, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, Freewheel, GDX Driver, Glue, Green Building Studio, Heads-up Design, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, iMOUT, Incinerator, Inferno, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, Map It, Build It, Use It, MatchMover, Maya, Mechanical Desktop, MIMI, Moldflow, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moondust, MotionBuilder, Movimento, MPA, MPA (design/logo), MPI (design/logo), MPX, MPX (design/logo), Mudbox, Multi-Master Editing, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, PolarSnap, Powered with Autodesk Technology, Productstream, ProMaterials, RasterDWG, RealDWG, Real-time Roto, Recognize, Render Queue, Retimer, Reveal, Revit, Revit LT, RiverCAD, Robot, Scaleform, Scaleform GFx, Showcase, Show Me, ShowMotion, SketchBook, Smoke, Softimage, Socialcam, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-

How to Contact Autodesk Scaleform:

Document	CLIK AS3 User Guide
Address	Autodesk Scaleform Corporation 6305 Ivy Lane, Suite 310 Greenbelt, MD 20770, USA
Website	www.scaleform.com
Email	info@scaleform.com
Direct	(301) 446-3200
Fax	(301) 446-3199

Table of Contents

1 简介 1

1.1 概要 1

1.1.1. 在 Scaleform CLIK 包含的内容 1

1.2 UI 关注 2

1.3 组件理解 3

1.3.1 Inspectable Properties 3

1.4 框架基础知识 4

1.4.1 事件 4

1.4.2 焦点 5

2 内置组件 6

2.1 基本按钮和文本类型 6

2.1.1 Button 8

2.1.2 CheckBox 12

2.1.3 Label 16

2.1.4 TextInput 18

2.1.5 TextArea 22

2.2 分组类型 26

2.2.1 RadioButton 26

2.2.2 ButtonGroup 31

2.2.3 ButtonBar 32

2.3 滚动类型 35

2.3.1 ScrollIndicator 35

2.3.2 ScrollBar 37

2.3.3 Slider 40

2.4 列表类型 42

2.4.1 NumericStepper 44

2.4.2 OptionStepper 46

2.4.3 ListItemRenderer 49

2.4.4 ScrollingList 53

2.4.5 TileList 58

2.4.6 DropdownMenu 63

2.5 进度类型 69

2.5.1 StatusIndicator 69

2.6 其他类型 71
2.6.1 Window 72
3 艺术细节 75
3.1 最优方法 75
3.1.1 像素图像 75
3.1.2 蒙版 76
3.1.3 动画 76
3.1.4 图层和绘制元素 76
3.1.5 复杂皮肤界面 77
3.1.6 2 的指数倍 77
3.2 已知问题和推荐工作流程 77
3.2.1 复制组件 77
3.3 皮肤绘制实例 79
3.3.1 StatusIndicator 皮肤绘制 80
3.4 字体和本地化 82
3.4.1 概要 82
3.4.2 内置字体 82
3.4.3 在 textField 嵌入字体 82
3.4.4 本地化系统 83
4 编程详述 84
4.1 UI 组件 UIComponent 84
4.1.1 初始化 85
4.2 组件状态 85
4.2.1 按钮组件 85
4.2.2 非按钮交互组件 87
4.2.3 非交互组件 87
4.2.4 特殊案例 87
4.3 事件模型 88
4.3.1 最佳使用方法 88
4.4 运行时创建组件 89
4.5 焦点处理 90
4.5.1 最佳使用方法 90
4.5.2 在复合附件捕获焦点 91
4.6 输入处理 91
4.6.1 最佳使用方法 91
4.6.2 多鼠标光标 93

4.7 失效 94
4.7.1 最佳使用方法 94
4.8 组件缩放 95
4.8.1 Scale9Grid 95
4.8.2 强制 95
4.9 组件和数据设置 96
4.9.1 最佳使用方法 96
4.10 动态动画 97
4.10.1 最佳使用方法 98
4.11 布局框架 98
4.11.1 布局 99
4.11.2 LayoutData 100
4.12 弹出式支持 101
4.12.1 最佳使用方法 101
4.13 拖放 102
4.13.1 最佳使用方法 102
5 实例 104
5.1 基础 104
5.1.1 包含两个 textField 的 ListItem Renderer 104
5.1.2 像素滚动视图 106
6 常见问题解答 108
7 CLIK AS3 vs. CLIK AS2 109

1

Scaleform® Common Lightweight Interface Kit CLIK™)
CLIK CLIK
Scaleform CLIK CLIK

1.1

Scaleform CLIK ActionScript™ 2.0 (AS2) Scaleform 4.0
PC
CPU UI
CLIK 15

1.1.1. Scaleform CLIK

UI

Button	Slider
ButtonBar	StatusIndicator
CheckBox	TileList
RadioButton	Label
TextInput	ScrollingList
TextArea	DropdownMenu
ScrollIndicator	NumericStepper
ScrollBar	

API

InputManager	UIComponent
FocusManager	Tween
DragManager	DataProvider
PopUpManager	IDataProvider
IUIComponent	IList
Constraints	IListItemRenderer

Getting Started with CLIK
Getting Started with CLIK Buttons
CLIK AS3 API Reference
CLIK AS3 User Guide
CLIK Flash Samples
CLIK Video Tutorials

1.2 UI

UI Microsoft Visio®
Flash UI Scaleform
CLIK

UI Flash Visio
Flash

UI

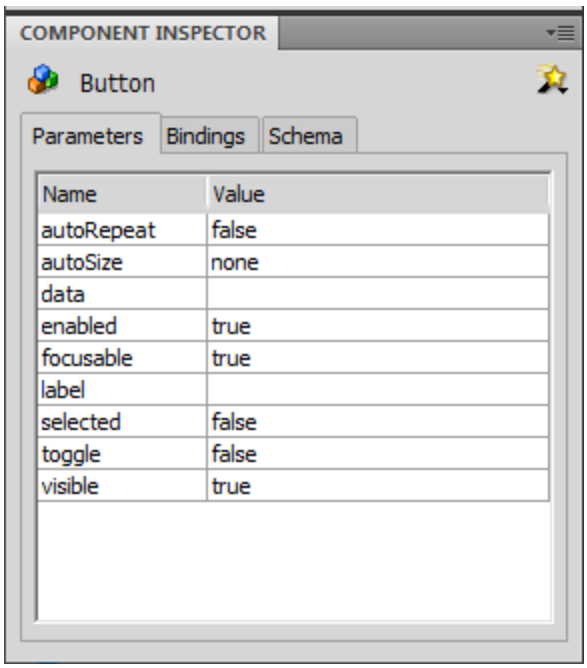
web

Flash
Flash

UI Flash

UI UI

Scaleform Flash ActionScript UI
Lua UnrealScript
Flash ActionScript Flash



SWF Flash IDE

CLIK

1.4

1.4.1

CLIK

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

ActionScript 3 Event Adobe ActionScript 3

http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/Event.html

“ ”

Scaleform Scaleform Extensions.enabled true Scaleform MouseEventEx MouseEvent MouseEvent MouseEventEx /
MouseEventEx

1.4.2

Flash stage
InteractiveObject

Programming CLIK
(Tab) (Shift+Tab)
CLIK
CLIK Flash CLIK CLIK
(Tab) (Shift+Tab)

Enter

Enter
Xbox360 PS3 A X UI CLIK

2

Scaleform CLIK UI CLIK

– – CLIK

CLIK UI

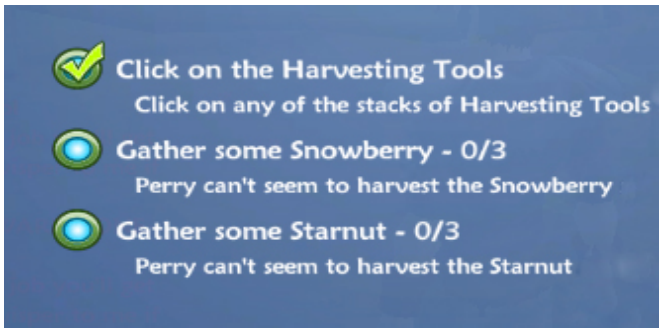
- ***User interaction***
- ***Component setup*** Flash
- ***States***
- ***Inspectable properties*** Flash
- ***Events*** UI
- ***Tips and tricks***

2.1

Button CheckBox Label TextInput TextArea Button

CheckBox Button Label TextInput TextArea

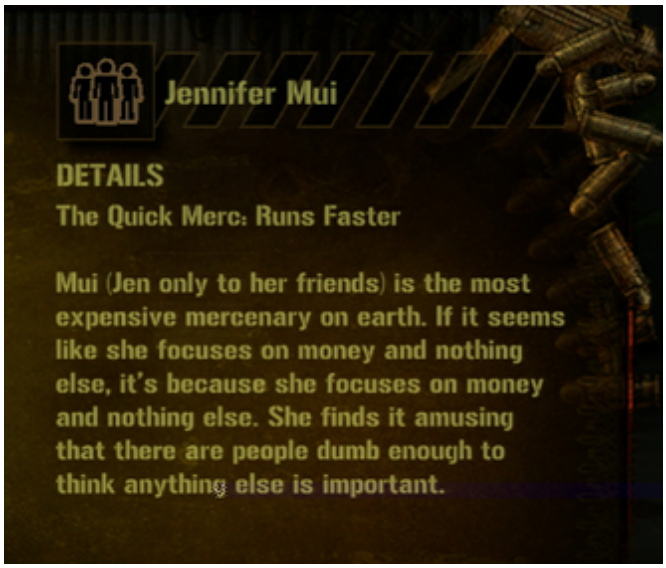
1 ***Free Realms***



2 ***Free Realms*** **Check box**

3 ***Free Realms*** **Label**

4: ***Crysis Warhead*** **Text input**



5 ***Mercenaries 2*** **Text area**

2.1.1 Button

6

Button CLIK
Button scaleform.clik.controls.Button textField
ScrollBar Slider

2.1.1.1

2.1.1.2

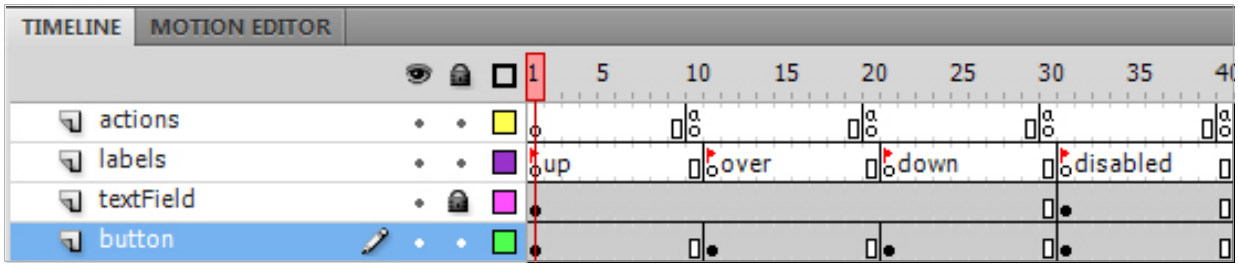
CLIK Button MovieClip

- ***textField***: ( ) TextField
- ***focusIndicator:*** ( ) MovieClip MovieClip
"show" "hide" over Button
over

2.1.1.3

CLIK

- ***up***
- ***over***
- ***down***
- ***disabled***

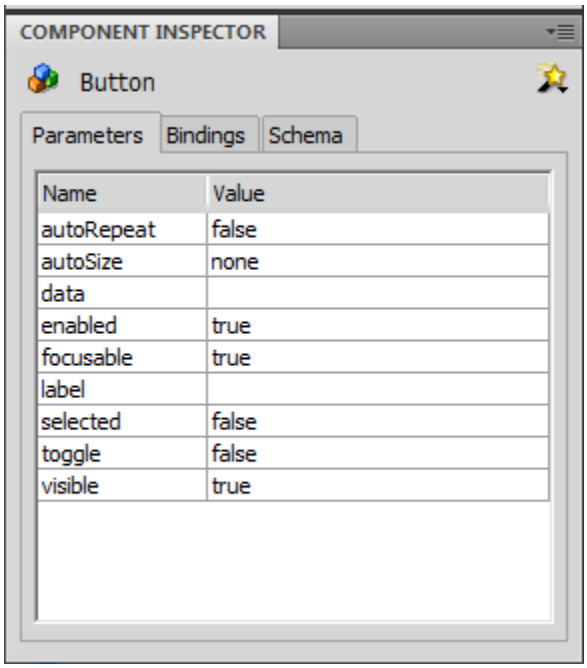


8 Button

Flash

CLIK

2.1.1.4



9 CS4

:

autoRepeat			"click"
autoSize		autoSize	TextFieldAutoSize.NONE
Data		ButtonGroup	
enabled	false		
focusable	/	focusable	false

	tab
Label	
selected	toggle true toggle false ActionScript
toggle	true
visible	false

2.1.1.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
Event.SELECT	
MouseEvent.ROLL_OVER	*mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx
MouseEvent.ROLL_OUT	*mouseIdx* uint Scaleform MouseEventEx

	buttonIdx Scaleform MouseEventEx
ButtonEvent.PRESS	*controllerIdx*: uint *isKeyboard* true - (Keyboard)/ (Gamepad) false – *isRepeat* true – autoRepeating false –
MouseEvent.DOUBLE_CLICK	*doubleClickEnabled* true *mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx
ButtonEvent.CLICK	*controllerIdx*: uint *isKeyboard* true - (Keyboard)/ (Gamepad) false – *isRepeat* true – autoRepeating false –
ButtonEvent.DRAG_OVER	*controllerIdx*: uint
ButtonEvent.DRAG_OUT	*controllerIdx*: uint
ButtonEvent.RELEASE_OUTSIDE	*controllerIdx*: uint

ActionScript

```
myButton.addEventListener( ButtonEvent.PRESS, onButtonPress );
function onButtonPress( e:ButtonEvent ):void {
      // Do something
}
```

“ButtonEvent.PRESS x” ‘myButton’ onButtonPress ‘e’

ButtonEvent

2.1.2 CheckBox

10 CheckBox

CheckBox (scaleform.clik.controls.CheckBox)
true/false ToggleButton Toggle

2.1.2.1

CheckBox

2.1.2.2

CLIK CheckBox MovieClip

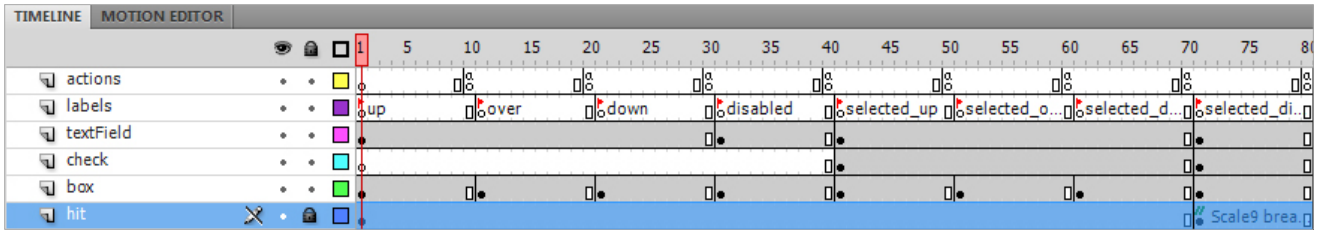
- ***textField***: ( ) TextField
- ***focusIndicator:*** ( ) MovieClip MovieClip
  MovieClip “show” “hide”

2.1.2.3

Toggle CheckBox

- ***up***
- ***over***
- ***down***
- ***disabled***
- ***selected_up***
- ***selected_over***

- *selected_ down*
- *selected_disabled*

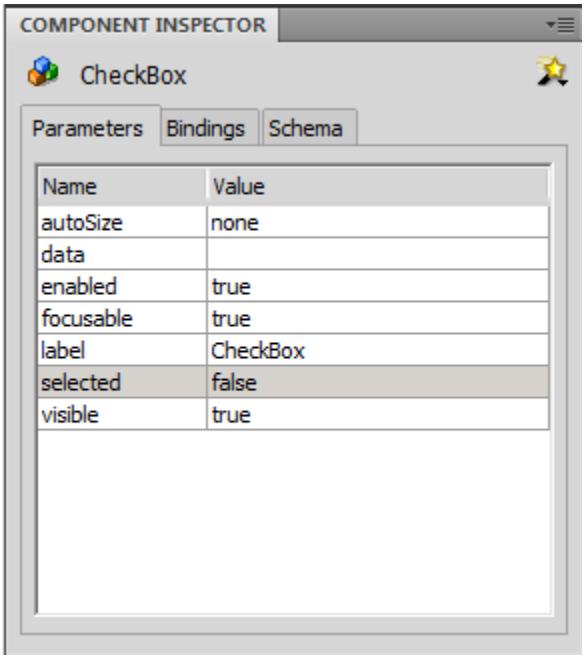


11: CheckBox

CheckBox Button

CLIK

2.1.2.4



CheckBox autoRepeat

Toggle

autoSize	autoSize	autoSize=TextFieldAutoSize.NONE	
Data	ButtonGroup		
Enabled	false		
focusable	/	focusable	false

	tab
Label	
selected	toggle true toggle false ActionScript
Visible	false

2.1.2.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

CheckBox

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
Event.SELECT	
MouseEvent.ROLL_OVER	*mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx
MouseEvent.ROLL_OUT	*mouseIdx* uint Scaleform MouseEventEx

	buttonIdx Scaleform MouseEventEx
ButtonEvent.PRESS	*controllerIdx*: uint *isKeyboard* true - (Keyboard)/ (Gamepad) false – *isRepeat* true – autoRepeating false –
MouseEvent.DOUBLE_CLICK	*doubleClickEnabled* true *mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx
ButtonEvent.CLICK	*controllerIdx*: uint *isKeyboard* true - (Keyboard)/ (Gamepad) false – *isRepeat* true – autoRepeating false –
ButtonEvent.DRAG_OVER	*controllerIdx*: uint
ButtonEvent.DRAG_OUT	*controllerIdx*: uint
ButtonEvent.RELEASE_OUTSIDE	*controllerIdx*: uint

CheckBox Toggle

```
myCheckBox.addEventListener(Event.SELECT, onCheckBoxToggle);
function onCheckBoxToggle(e:Event):void {
      // Do something
}
```

2.1.3 Label

12:

CLIK Label (scaleform.clik.controls.Label) textField MovieClip
Label textField

textField textField

Label MovieClip textField
textField Label disabled
textField AS2

Label Label
textField Label
textField Label

2.1.3.1

Label

2.1.3.2

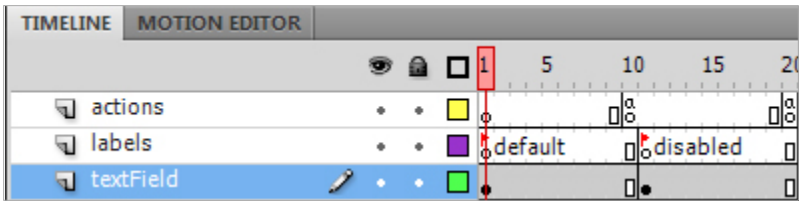
CLIK Label MovieClip

- ***textField***: TextField Label

2.1.3.3

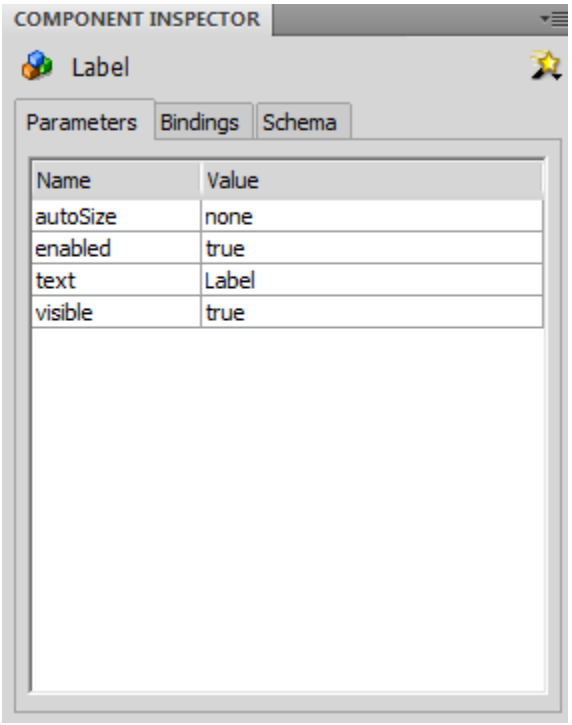
CLIK Label

- ***default*** enabled
- ***disabled***



13: Label

2.1.3.4



Label

text		
visible	false	
enabled	false	
autoSize	autoSize	autoSize=TextFieldAutoSize.NONE

2.1.3.5

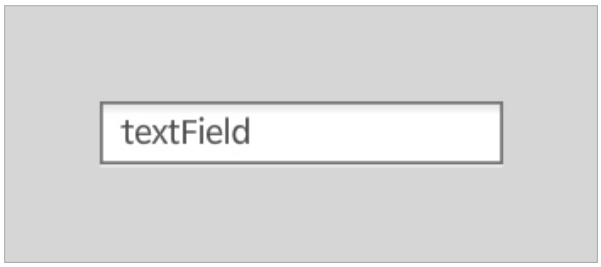
Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

Label

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	

2.1.4 TextInput

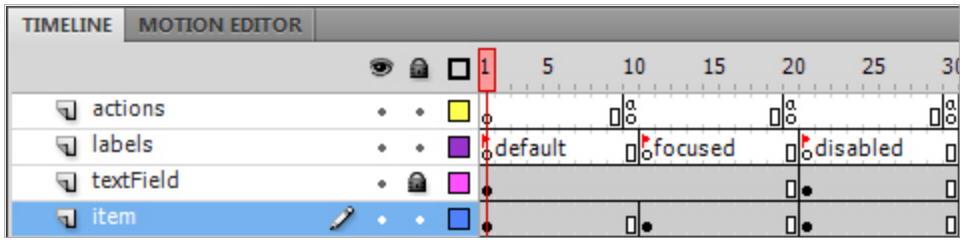


14: TextInput.

TextInput (scaleform.clik.controls.TextInput) textField
Label textField textField
HTML TextInput
textField

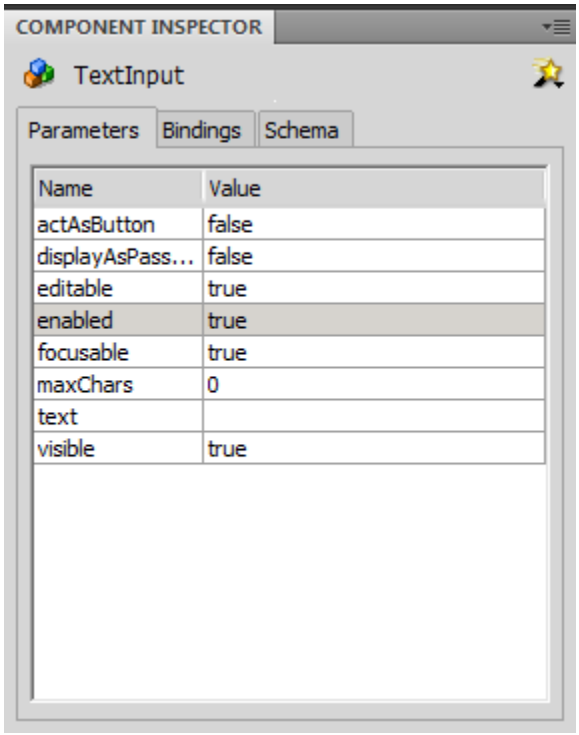
textField Label Label
textFields TextInput UI
PC TextInput textField

TextInput disabled textField
TextInput TextInput AS2



15: TextInput

2.1.4.4



TextInput :

actAsButton	" " " " tab " "
displayAsPassword	true textField '*' textField
editable	false TextInput
enabled	false
focusable	/ focusable false tab
maxChars	textField

text	textField
visible	false

2.1.4.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

TextInput

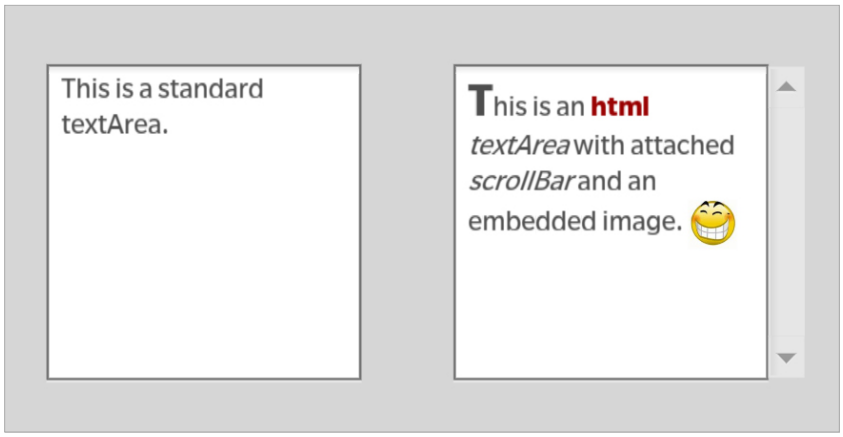
ComponentEvent.SHOW	visible	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
Event.SELECT		
MouseEvent.ROLL_OVER	*mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform	MouseEventEx
MouseEvent.ROLL_OUT	*mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform	MouseEventEx

textField

```
myTextInput.addEventListener(Event.CHANGE, onTextChange);
```

```
function onTextChange(e:Event):void {
      trace("Latest text: " + e.target.text);
}
```

2.1.5 TextArea



16: TextArea.

TextArea (scaleform.clik.controls.TextArea) CLIK TextInput
ScrollBar " "
" " " "

Label TextInput TextArea textField textField
HTML textField
TextArea

textField ScrollIndicator ScrollBar TextArea
textField TextArea
TextArea

2.1.5.1

TextArea textField
textField

2.1.5.2

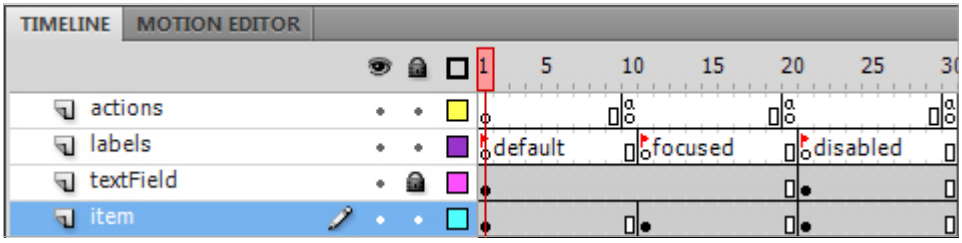
CLIK TextArea MovieClip

- ***textField***: TextField

2.1.5.3

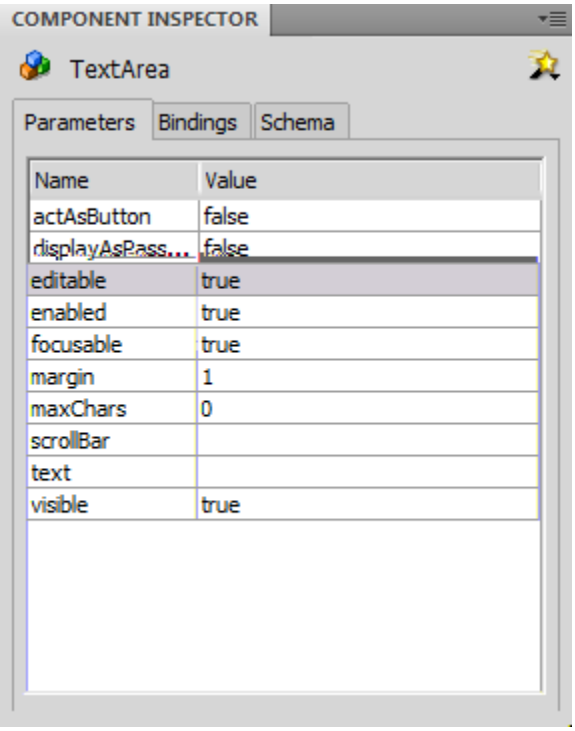
TextInput TextArea disabled

- ***default*** enabled
- ***focused*** textField
- ***disabled***



17: TextArea

2.1.5.4



TextArea TextInput CLIK ScrollBar
2.4

actAsButton	true TextInput rollOver rollOut tab TextInput
displayAsPassword	true textField '*' textField
editable	false TextInput
enabled	false
focusable	/ focusable false tab
maxChars	textField
scrollBar	CLIK ScrollBar ScrollBar ID TextArea
scrollPolicy	"auto" "on" ScrollBar "off" ScrollBar ScrollBar
text	textField
visible	false

2.1.5.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

TextArea

ComponentEvent.SHOW	visible true
ComponentEvent.HIDE	visible false
FocusHandlerEvent.FOCUS_IN	
Event.CHANGE	textField .
Event.SCROLL	textField
ComponentEvent.STATE_CHANGE	
MouseEvent.ROLL_OVER	*mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx
MouseEvent.ROLL_OUT	*mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx

TextArea

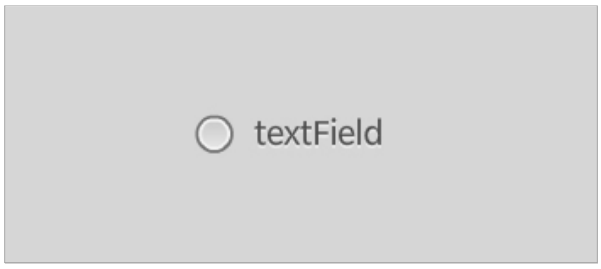
```
myTextArea.addEventListener(Event.SCROLL, onTextScroll);
function onTextScroll(e:Event):void {
	// Do something
}
```

2.2

RadioButton ButtonGroup ButtonBar ButtonGroup
Button ButtonBar
RadioButton ButtonGroup

18: ***Dawn of War II***

2.2.1 RadioButton



19: **RadioButton**

RadioButton (scaleform.clik.controls.RadioButton)

CLIK CheckBox
ButtonGroup

2.2.1.1

2.2.1.2

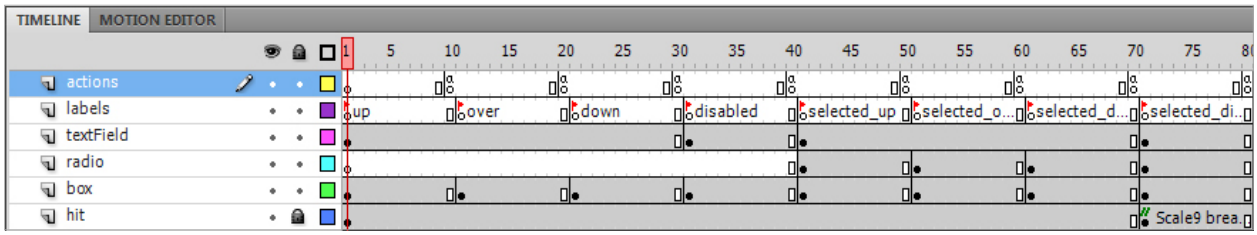
CLIK RadioButton MovieClip

- ***textField*** ( ) TextField
- ***focusIndicator*** MovieClip MovieClip
  MovieClip “show” “hide”

2.2.1.3

CheckBox

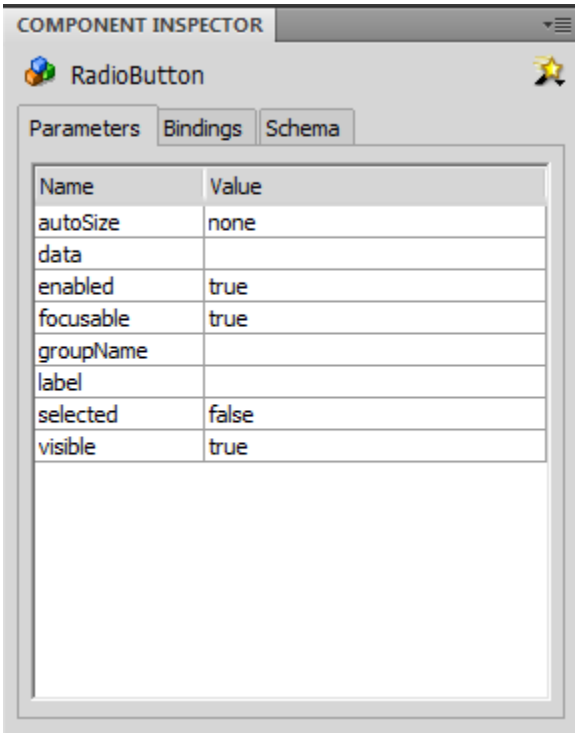
- ***up***
- ***over***
- ***down***
- ***disabled***
- ***selected_up***
- ***selected_over***
- ***selected_ down***
- ***selected_disabled***



20: RadioButton

RadioButton Button
RadioButton CLIK

2.2.1.4



Button RadioButton
disableFocus Toggle

autoSize	autoSize autoSize=TextFieldAutoSize.NONE
data	ButtonGroup
enabled	false
focusable	/ focusable false tab
groupName	ButtonGroup RadioButton RadioButton ButtonGroup RadioButton RadioButton _root _root ButtonGroup RadioButton
label	
selected	toggle true toggle false ActionScript

2.2.1.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

RadioButton

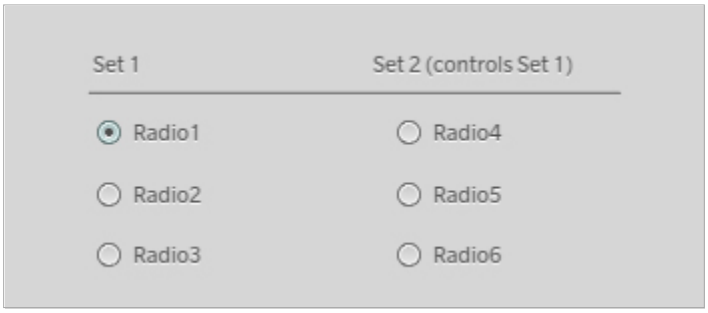
ComponentEvent.SHOW		true
ComponentEvent.HIDE		false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
Event.SELECT		
MouseEvent.ROLL_OVER	*mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx	
MouseEvent.ROLL_OUT	*mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx	
ButtonEvent.PRESS	*controllerIdx*: uint *isKeyboard* true - (Keyboard)/ (Gamepad) false – *isRepeat* true – autoRepeating false –	
MouseEvent.DOUBLE_CLICK		*doubleClickEnabled* true

	mouseIdx uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx
ButtonEvent.CLICK	*controllerIdx*: uint *isKeyboard* true - (Keyboard)/ (Gamepad) false – *isRepeat* true – autoRepeating false –
ButtonEvent.DRAG_OVER	*controllerIdx*: uint
ButtonEvent.DRAG_OUT	*controllerIdx*: uint
ButtonEvent.RELEASE_OUTSIDE	*controllerIdx*: uint

RadioButton

```
myRadio.addEventListener(Event.SELECT, onRadioToggle);
function onRadioToggle(e:Event):void {
      // Do something
}
```

2.2.2 ButtonGroup



21: ButtonGroup.

CLIK ButtonGroup (scaleform.clik.controls.ButtonGroup)

CLIK CheckBox RadioButton
ButtonGroup

2.2.2.1

RadioButton

2.2.2.2

CLIK ButtonGroup MovieClip

2.2.2.3

ButtonGroup

2.2.2.4

ButtonGroup

2.2.2.5

Event Event

- ***type***
- ***target***

- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

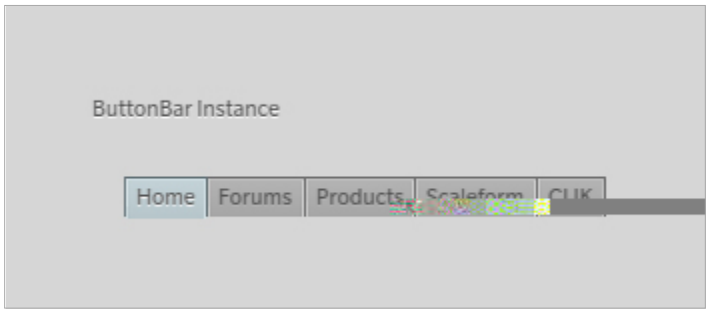
ButtonGroup

Event.CHANGE
ButtonEvent.CLICK
target: CLIK Button

ButtonGroup

```
myGroup.addEventListener(Event.CHANGE, onNewSelection);
function onNewSelection(e:Event):void {
	if (e.item == myRadio) {
		// Do something
	}
}
```

2.2.3 ButtonBar



22: ButtonBar.

ButtonBar (scaleform.clik.controls.ButtonBar) ButtonGroup
dataProvider dataProvider
tab UI

```
buttonBar.dataProvider = ["item1", "item2", "item3", "item4", "item5"];
```

2.2.3.1

ButtonGroup

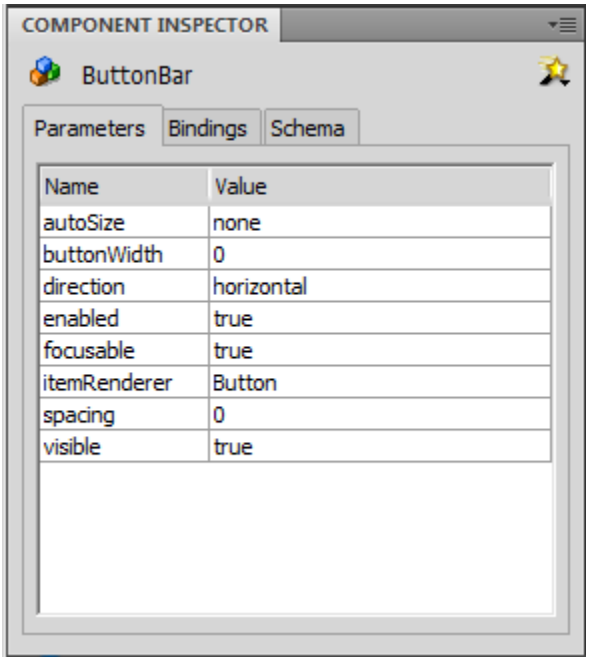
2.2.3.2

CLIK ButtonBar MovieClip

2.2.3.3

CLIK

2.2.3.4



Flash IDE

ButtonBar

autoSize	true Button
Direction	“ ”(Horizontal) Button “ ” (Vertical)
buttonWidth	Button autoSize true
Enabled	false
Focusable	/ focusable false

	tab
itemRenderer	ID
Spacing	*direction*
Visible	false ButtonBar

2.2.3.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

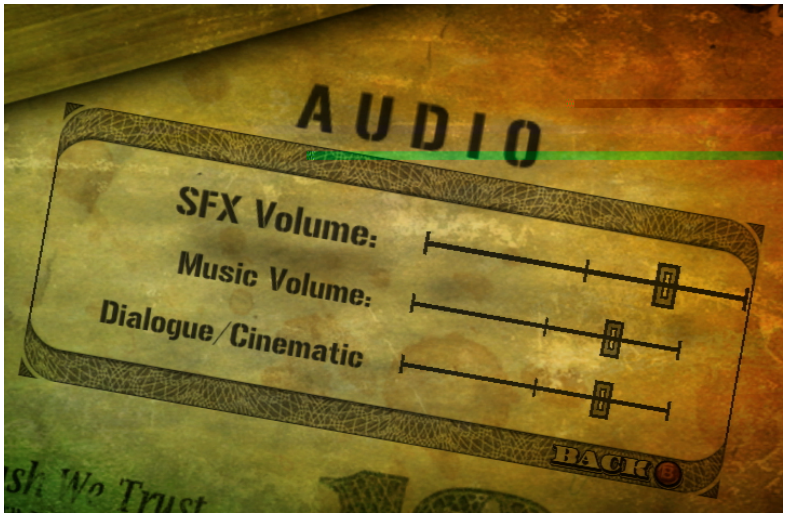
ButtonBar

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
IndexEvent.INDEX_CHANGE	*index* ButtonBar int -1 1 *lastIndex* ButtonBar int -1 1 *data* dataProvider AS3 Object
ButtonBarEvent.BUTTON_SELECT	*index* ButtonBar Int -1 1 *renderer* ButotnBar Button Button

```
myBar.addEventListener(ButtonBarEvent.BUTTON_SELECT, onItemClick);
function onItemClick(e:ButtonBarEvent) {
      processData(e.renderer.data);
      // Do something
}
```

2.3

ScrollIndicator ScrollBar Slider ScrollIndicator
ScrollBar
Slider



23: ***Mercenaries 2***

2.3.1 ScrollIndicator

24: **ScrollIndicator.**

CLIK ScrollIndicator (scaleform.clik.controls.ScrollIndicator)
textField TextArea
ID

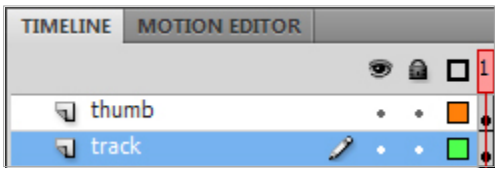
2.3.1.1

2.3.1.2

CLIK ScrollIndicator MovieClip

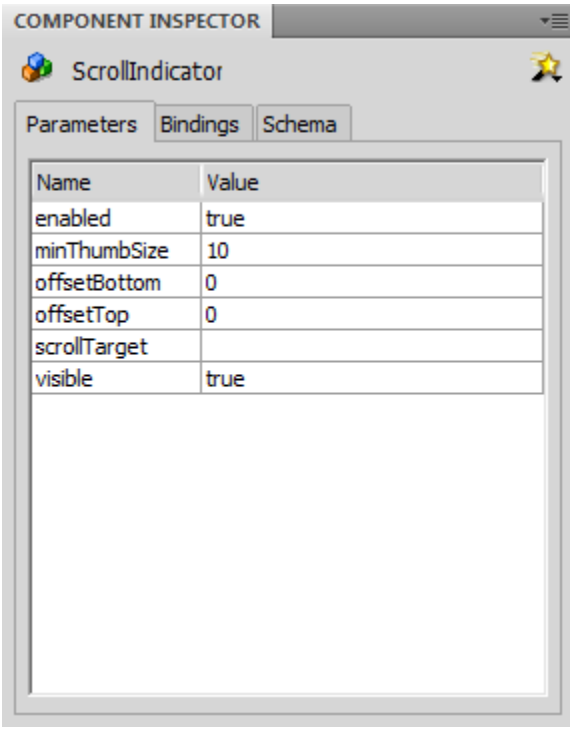
- ***thumb***: CLIK Button
- ***track*** MovieClip

2.3.1.3



25: ScrollIndicator

2.3.1.4



Enabled	false			
offsetTop				
offsetBottom				
scrollTarget	TextArea	textField	textField	ScrollIndicator
Visible	false			

2.3.1.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

ScrollIndicator

ComponentEvent.SHOW	visible	true	
ComponentEvent.HIDE	visible	false	
Event.SCROLL	ScrollIndicator		

```
mySI.addEventListener(Event.SCROLL, onTextScroll);
function onTextScroll(e:Event) {
	trace("mySI.position: " + e.target.position);
	// Do something
}
```

2.3.2 ScrollBar

26: ScrollBar.

CLIK ScrollBar(scaleform.clik.controls.ScrollBar)

2.3.2.1

ScrollBar

trackMode

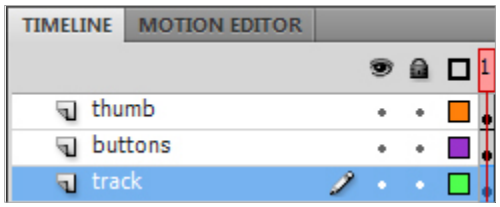
trackMode (Shift)

2.3.2.2

CLIK ScrollBar MovieClip

- ***upArrow*** CLIK
- ***downArrow***: CLIK
- ***thumb***: CLIK
- ***track***: CLIK

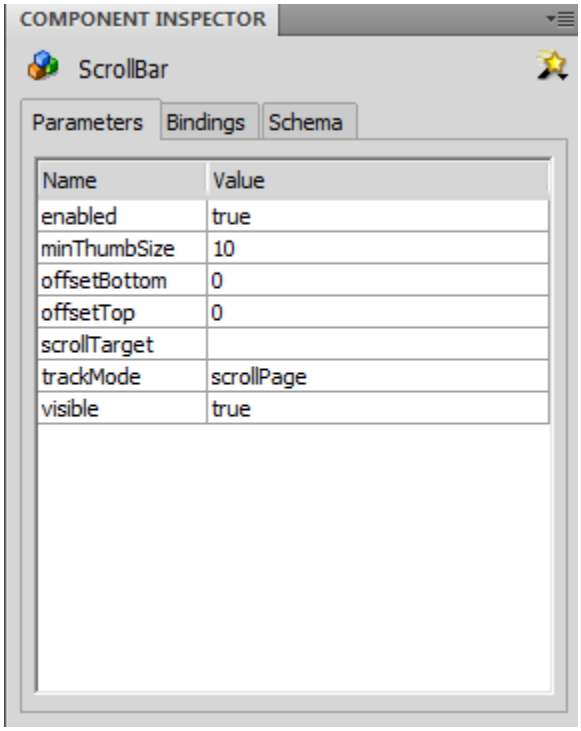
2.3.2.3



27: ScrollBar

SrollBar ScrollIndicator

2.3.2.4



Enabled	false		
offsetTop			
offsetBottom			
scrollTarget		TextArea	textField
trackMode	scrollToCursor	scrollPage	
Visible	false		

2.3.2.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event

- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

ScrollBar

ComponentEvent.SHOW	visible		true
ComponentEvent.HIDE	visible	false	
Event.SCROLL	ScrollIndicator		

```
mySB.addEventListener(Event.SCROLL, onTextScroll);
function onTextScroll(e:Event) {
	trace("mySB.position: " + e.target.position);
	// Do something
}
```

2.3.3 Slider

28: Slider.

Slider(scaleform.clik.controls.Slider)

2.3.3.1

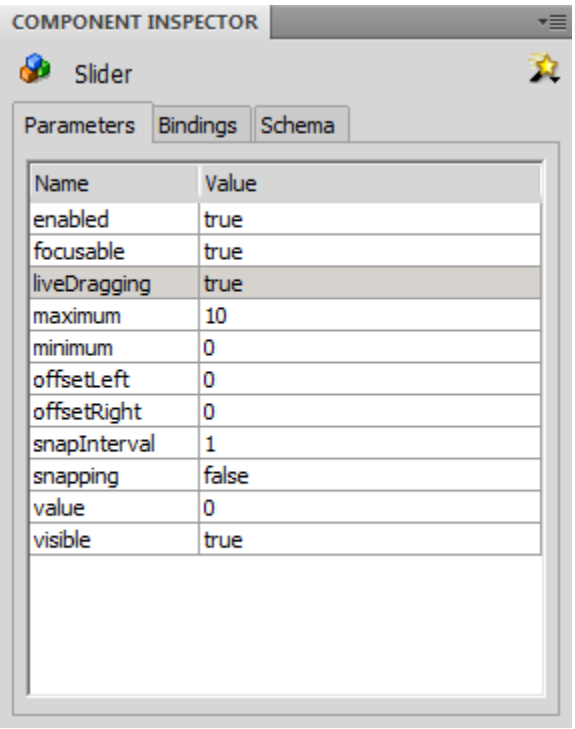
home end

2.3.3.2

CLIK Slider MovieClip

- ***thumb*** CLIK
- ***Track*** CLIK

2.3.3.3



Enabled	false
Focusable	/ focusable false tab
Value	Slider
minimum	
maximum	
snapping	true
snapInterval	false
liveDragging	true Slider
offsetLeft	

offsetRight	
Visible	false

2.3.3.4

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

Slider

ComponentEvent.SHOW	true
ComponentEvent.HIDE	false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
SliderEvent.VALUE_CHANGE	Slider's

Slider

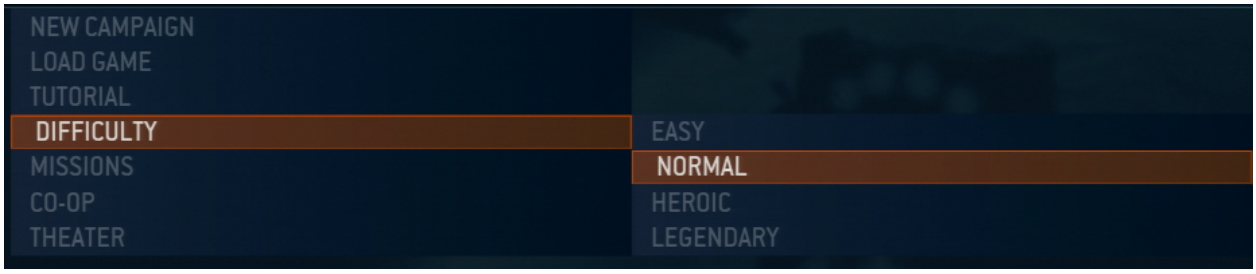
```
mySlider.addEventListener(SliderEvent.VALUE_CHANGE, onValueChange);
function onValueChange(e:SliderEvent) {
      trace("mySlider.value: " + e.target.value);
      // Do something
}
```

2.4

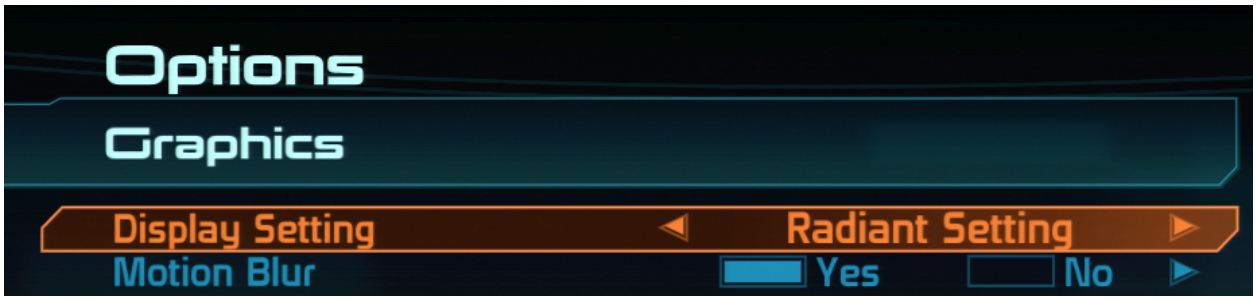
CLIK NumericStepper OptionStepper ScrollingList TileList DropdownMenu NumericStepper DataProvider ListItemRenderer ScrollingList TileList

NumericStepper OptionStepper ScrollingList TileList
ScrollIndicator ScrollBar DropdownMenu
ScrollingList TileList

30: ***Mercenaries 2***



31: ***Halo Wars*** **'Difficulty Settings'**



32: ***Mass Effect*** **'Display Setting'**

2.4.1 NumericStepper

NumericStepper (scaleform.clik.controls.NumericStepper)

2.4.1.1

NumericStepper

(Home)

(End)

2.4.1.2

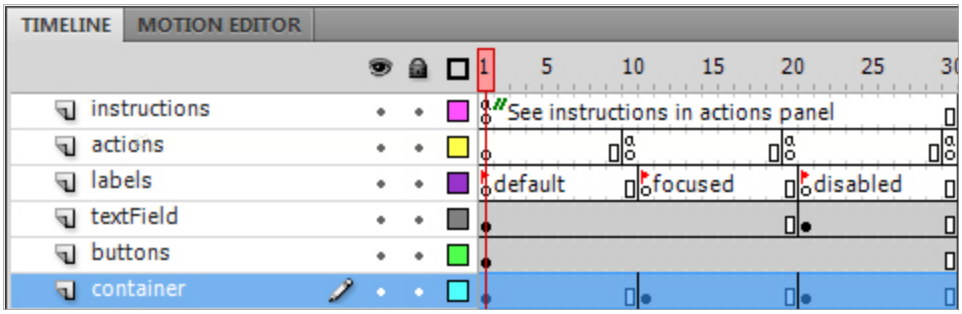
CLIK NumericStepper MovieClip

- ***textField***: TextField
- ***nextBtn***: CLIK Button
- ***prevBtn***: CLIK Button

2.4.1.3

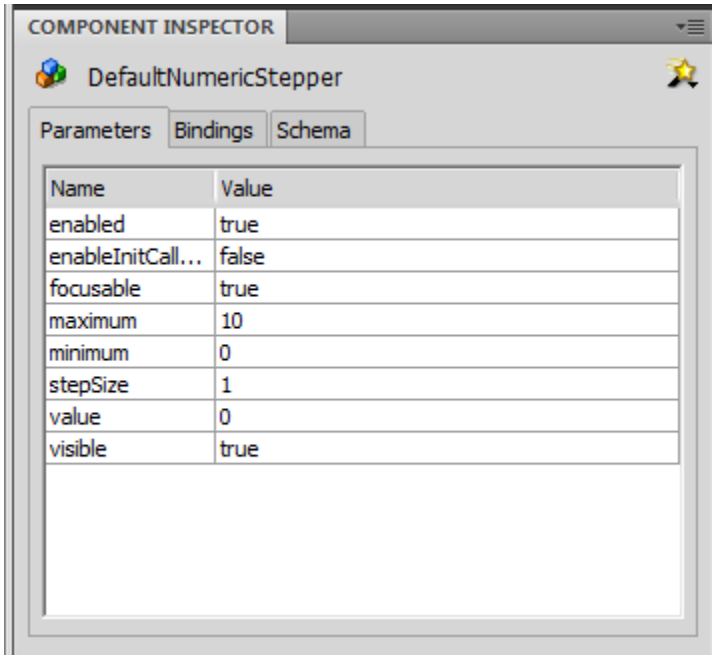
NumericStepper

- ***default*** enabled
- ***focused*** textField
- ***disabled***



34: NumericStepper

2.4.1.4



NumericStepper MovieClip

Enabled	false
Focusable	/ focusable false tab
minimum	NumericStepper
maximum	NumericStepper
stepSize	NumericStepper / NumericStepper
Value	NumericStepper.
Visible	false

2.4.1.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

NumericStepper

ComponentEvent.SHOW	visible true
ComponentEvent.HIDE	visible false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN	
FocusHandlerEvent.FOCUS_OUT	
ComponentEvent.STATE_CHANGE	
IndexEvent.INDEX_CHANGE	NumericStepper *index* NumericStepper int -1 1 *lastIndex* NumericStepper int -1 1 *data* dataProvider AS3 Object

NumericStepper

```
myNS.addEventListener(IndexEvent.INDEX_CHANGE, onValueChange);
function onValueChange(e:IndexEvent) {
	trace("myNS.value: " + e.target.value);
	// Do something
}
```

2.4.2 OptionStepper

35: OptionStepper.

OptionStepper (scaleform.clik.controls.OptionStepper), NumericStepper
dataProvider
OptionStepper
dataProvider

```
      optionStepper.dataProvider = new DataProvider(["item1", "item2", "item3",
"item4"]);
```

2.4.2.1

NumericStepper OptionStepper

(Home) (End) dataProvider

2.4.2.2

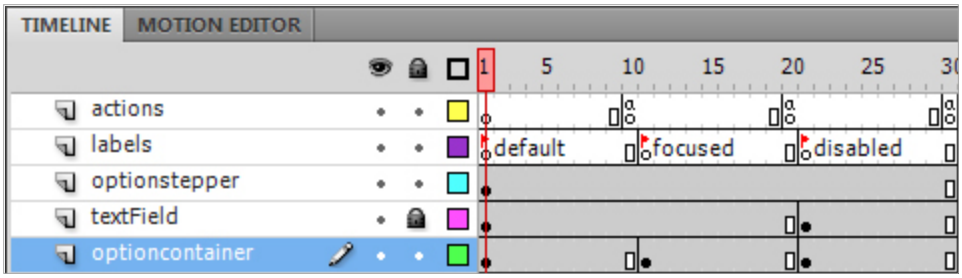
CLIK NumericStepper MovieClip :

- ***textField*** TextField
- ***nextBtn*** CLIK Button dataProvider
- ***prevBtn*** CLIK Button dataProvider

2.4.2.3

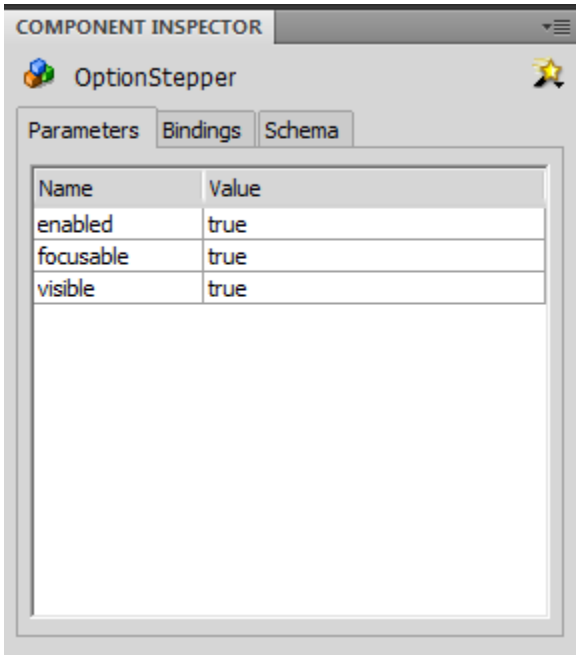
OptionStepper

- ***default*** enabled
- ***focused*** textField
- ***disabled***



36: OptionStepper

2.4.2.4



OptionStepper MovieClip

Enabled	false
Focusable	/ focusable false tab
Visible	false

2.4.2.5

Event Event

- ***type***
- ***target***

- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

OptionStepper

ComponentEvent.SHOW	visible		true	
ComponentEvent.HIDE	visible		false	
ComponentEvent.STATE_CHANGE				
FocusHandlerEvent.FOCUS_IN				
FocusHandlerEvent.FOCUS_OUT				
IndexEvent.INDEX_CHANGE	OptionStepper			
	index	NumericStepper	int	-1
		1		

ListItemRenderer (scaleform.clik.controls.ListItemRenderer) CLIK Button

ScrollingList TitleList DropdownMenu

2.4.3.1

ListItemRenderer Button
ListItemRenderer
ListItemRenderer

2.4.3.2

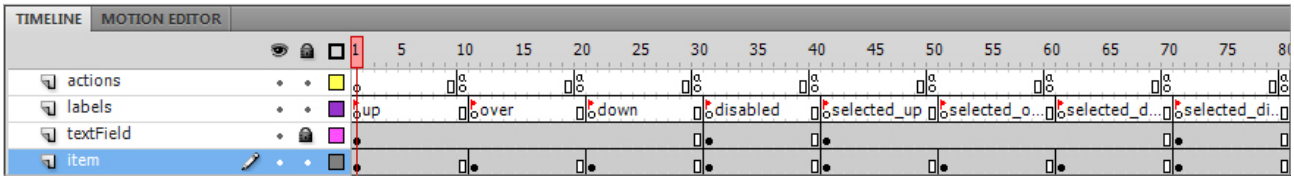
CLIK ListItemRenderert MovieClip :

- ***textField*** ( ) TextField
- ***focusIndicator***: ( ) MovieClip MovieClip
  “show” “hide”

2.4.3.3

ListItemRenderer *selected*

- ***up***
- ***over***
- ***down***
- ***disabled***
- ***selected_up***
- ***selected_over***
- ***selected_ down***
- ***selected_disabled***

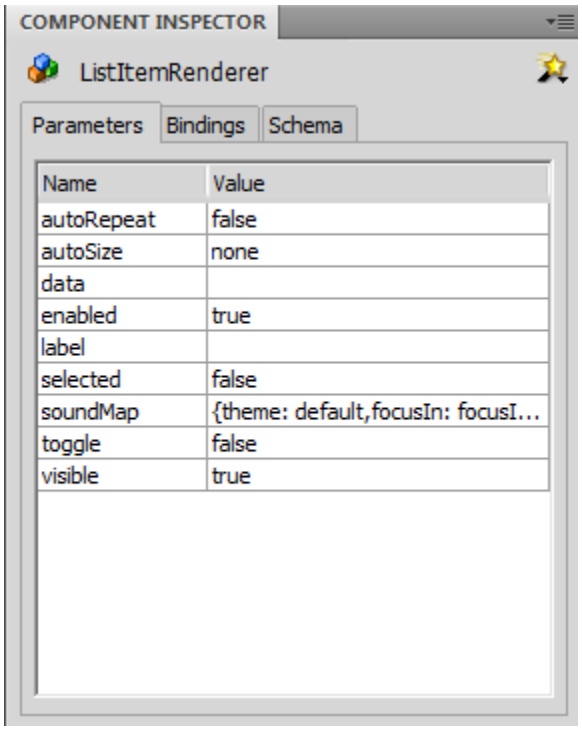


38: ListItemRenderer

ListItemRenderer Button

ListItemRenderer CLIK

2.4.3.4



ListItemRendere

autoRepeat	ListItemRender "click"
autoSize	ListItemRender ListItemRender autoSize autoSize=TextFieldAutoSize.NONE
Data	ListItemRender List (ScrollingList / TileList) ListItemRender
Enabled	false
Label	ListItemRenderer
selected	ListItemRender ListItemRender ListItemRender toggle true toggle false ActionScript
Toggle	ListItemRender toggle true ListItemRender

Visible	false

2.4.3.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

ListItemRenderer

ListItemRenderer's List ListItemRenderer ITEM_PRESS ITEM_CLICK ListEvents EventListener (ListEvent.ITEM_CLICK) ListItemRenderer EventListener

ComponentEvent.SHOW	visible	true	
ComponentEvent.HIDE	visible	false	
ComponentEvent.STATE_CHANGE			
FocusHandlerEvent.FOCUS_IN			
FocusHandlerEvent.FOCUS_OUT			
Event.SELECT			
MouseEvent.ROLL_OVER	*mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx		
MouseEvent.ROLL_OUT	*mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx		
ButtonEvent.PRESS	*controllerIdx*:		

	uint *isKeyboard* true - (Keyboard)/ (Gamepad) false – *isRepeat* true – autoRepeating false –
MouseEvent.DOUBLE_CLICK	*doubleClickEnabled* true *mouseIdx* uint Scaleform MouseEventEx *buttonIdx* Scaleform MouseEventEx
ButtonEvent.CLICK	*controllerIdx*: uint *isKeyboard* true - (Keyboard)/ (Gamepad) false – *isRepeat* true – autoRepeating false –
ButtonEvent.DRAG_OVER	*controllerIdx*: uint
ButtonEvent.DRAG_OUT	*controllerIdx*: uint
ButtonEvent.RELEASE_OUTSIDE	*controllerIdx*: uint

2.4.4 ScrollingList

39: ScrollingList.

ScrollingList (scaleform.clik.controls.ScrollingList)

ScrollIndicator ScrollBar

dataProvider

```
      scrollingList.dataProvider = new DataProvider(["item1", "item2", "item3",
"item4"]);
```

crollingList ListItemRenderer

FLA

2.4.4.1

ScrollBar ScrollingList

ScrollingList

ScrollingList *wrapping* *wrapping*

“normal” *wrapping* “wrap”

wrapping “stick”

(Page Up) (Page down)

(Home) (End) ScrollBar

ScrollingList ScrollaBar

D-Pad

CLIK UI

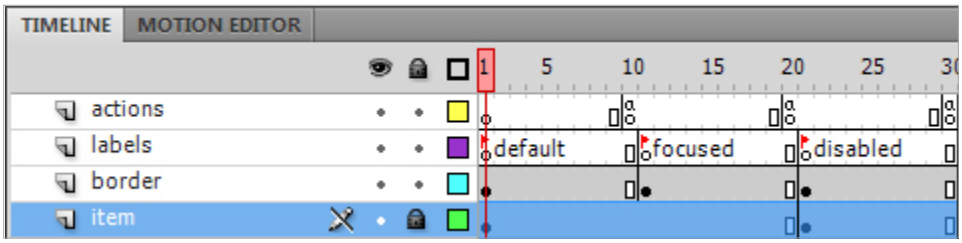
2.4.4.2

ScrollingList ScrollingList

2.4.4.3

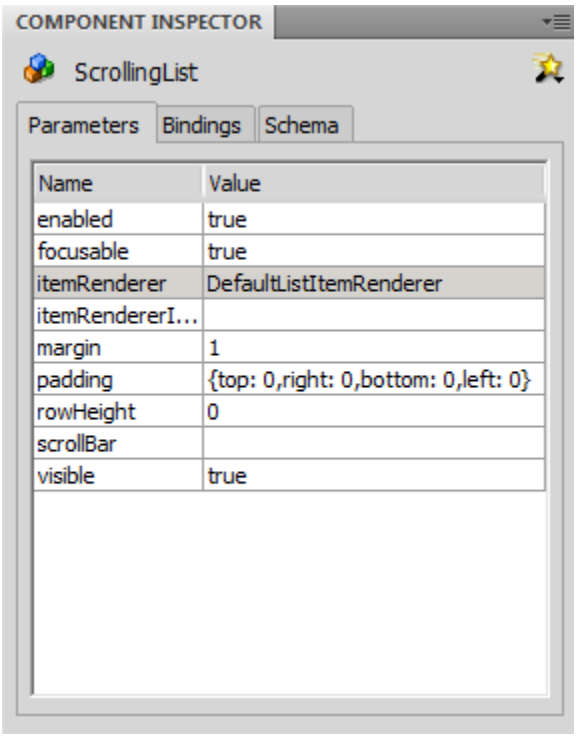
ScrollingList

- ***default*** enable
- ***focused***
- ***disabled***



40: ScrollingList

2.4.4.4



ScrollingList MovieClip

Enabled	false		
Focusable	/ tab	focusable	false
itemRenderer	ListItemRenderer *itemRendererInstanceName*		ListItemRenderers

itemRendererInstanceName	ScrollingList 'r' 'r1' 'r2' 'r3'….. 1
Margin	*rendererInstanceName*
Padding	(Padding) itemRendererInstanceName ListItemRenderer ScrollBar
rowHeight	*itemRendererInstanceName* ListItemRenderer
Visible	false ScrollBar ListItemRenderer

2.4.4.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

ScrollingList

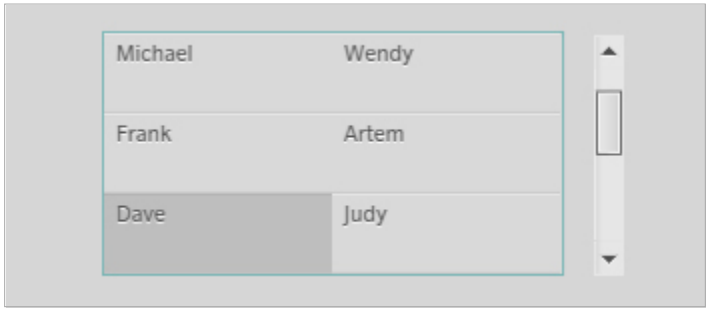
ComponentEvent.SHOW	visible true
ComponentEvent.HIDE	visible false
ComponentEvent.STATE_CHANGE	
FocusHandlerEvent.FOCUS_IN **FocusHandlerEvent.FOCUS_OUT**	
ListEvent.INDEX_CHANGE	• *itemRenderer* IListItemRenderer • *itemData*

	dataProvider	ActionScript Object	
	• *index*	int	-1 1
	• *controllerIdx*		/
ListEvent.ITEM_PRESS			
	• *itemRenderer*	IListItemRenderer	
	• *itemData* dataProvider	ActionScript Object	
	• *index*	int	-1 1
	• *controllerIdx*		/
ListEvent.ITEM_CLICK			
	• *itemRenderer*	IListItemRenderer	
	• *itemData* dataProvider	ActionScript Object	
	• *index*	int	-1 1
	• *controllerIdx*		/
ListEvent.ITEM_DOUBLE_CLICK			
	• *itemRenderer*	IListItemRenderer	
	• *itemData* dataProvider	ActionScript Object	
	• *index*	int	-1 1
	• *controllerIdx*		/
ListEvent.ITEM_ROLL_OVER			
	• *itemRenderer*	IListItemRenderer	
	• *itemData*		

	dataProvider • *index* • *controllerIdx*	ActionScript Object int -1 1 /
ListEvent.ITEM_ROLL_OUT	• *itemRenderer* • *itemData* dataProvider • *index* • *controllerIdx*	IListItemRenderer ActionScript Object int -1 1 /

```
myList.addEventListener(ListEvent.ITEM_CLICK, onItemClicked);
function onItemClicked(e:ListEvent) {
	trace("ListItemRenderer Clicked: " + e.itemRenderer);
	// Do something
}
```

2.4.5 TileList



41: TileList.

TileList (scaleform.clik.controls.TileList) ScrollingList

ScrollIndicator

ScrollBar TileList ScrollingList

dataProvider

```
      tileList.dataProvider = new DataProvider(["item1", "item2", "item3",
"item4", "item5"]);
```

TileList ListItemRenderer

FLA

2.4.5.1

ScrollBar TileList

TileList

(Page Up) (Page down)
(Home) (End) ScrollBar
TileList ScrollBar

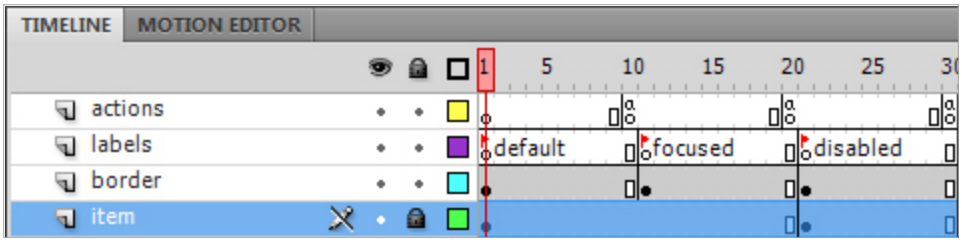
2.4.5.2

TileList TileList

2.4.5.3

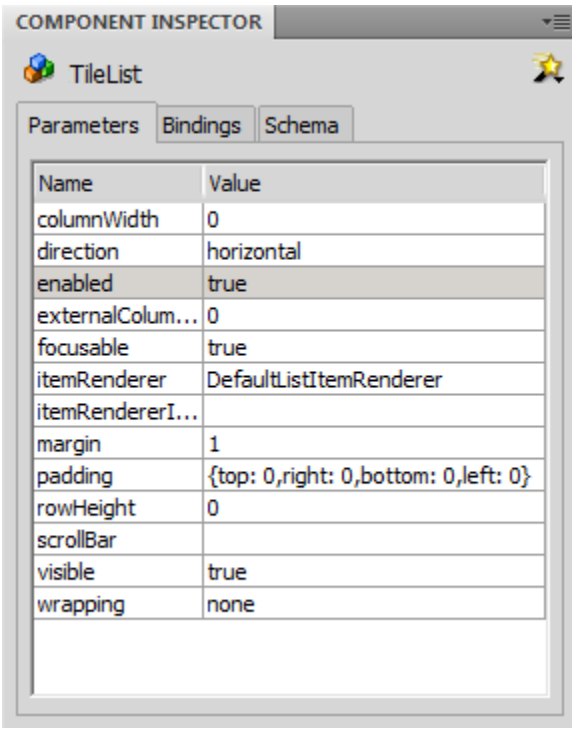
TileList

- ***default*** enable
- ***focused***
- ***disabled***



42: TileList

2.4.5.4



TileList MovieClip

columnWidth	*rendererInstanceName*
Direction	
Enabled	false
externalColumnCount	*rendererInstanceName* TileList
Focusable	/ focusable false tab
itemRenderer	ListItemRenderer

	itemRendererInstanceName	ListItemRenderer
itemRendererInstanceName	ScrollingList	Stage 'r' 'r1', 'r2', 'r3',… 1
Margin	*itemRendererInstanceName*	ListItemRenderer
Padding	(Padding) itemRendererInstanceName	ListItemRenderer ScrollBar
rowHeight	ListItemRenderer	*itemRendererInstanceName*
Visible	false ListItemRenderer	ScrollBar

2.4.5.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

TileList

ComponentEvent.SHOW	visible	true
ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
ListEvent.INDEX_CHANGE	• *itemRenderer*	IListItemRenderer

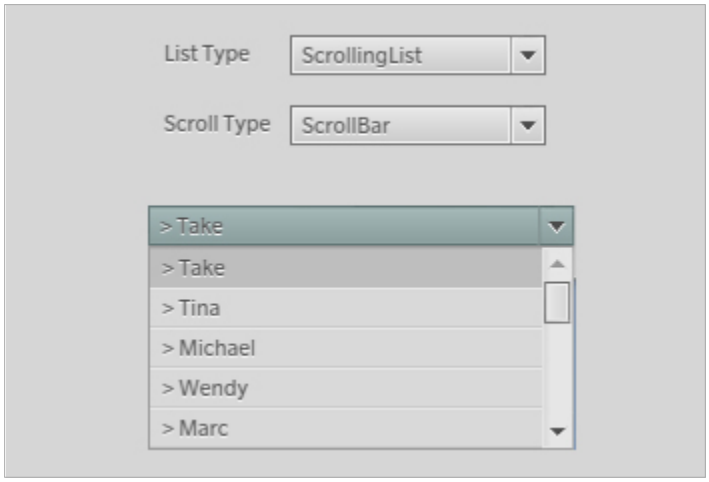
	• *itemData* dataProvider • *index* • *controllerIdx*	ActionScript Object int -1 1 /
	• *itemRenderer* • *itemData* dataProvider • *index* • *controllerIdx*	IListItemRenderer ActionScript Object int -1 1 /
	• *itemRenderer* • *itemData* dataProvider • *index* • *controllerIdx*	IListItemRenderer ActionScript Object int -1 1 /
	• *itemRenderer* • *itemData* dataProvider • *index* • *controllerIdx*	IListItemRenderer ActionScript Object int -1 1 /
	• *itemRenderer*	IListItemRenderer

	• *itemData* dataProvider	ActionScript Object
	• *index*	int -1 1
	• *controllerIdx*	/
ListEvent.ITEM_ROLL_OUT	• *itemRenderer*	IListItemRenderer
	• *itemData* dataProvider	ActionScript Object
	• *index*	int -1 1
	• *controllerIdx*	/

TileList

```
myList.addEventListener(FocusHandlerEvent.FOCUS_IN, onListFocused);
function onListFocused(e:FocusHandlerEvent) {
      trace("myList is now focused!”);
      // Do something
}
```

2.4.6 DropdownMenu



43: DropdownMenu.

DropdownMenu (scaleform.clik.controls.DropdownMenu)
DropdownMenu
ScrollingList TileList ScrollBar ScrollIndicator
dataProvider DropdownMenu dataProvide
dataProvide

```
dropdownMenu.dataProvider = new DataProvider(["item1", "item2", "item3",
"item4"]);
```

DropdownMenu ListItemRenderer
DropdownMenu ListItemRenderer FLA

DropdownMenu ScrollBar
ScrollIndicator DropdownMenu

2.4.6.1

DropdownMenu (Enter)
ScrollingList TileList ScrollBar
DropdownMenu

2.4.6.2

DropdownMenu MovieClip

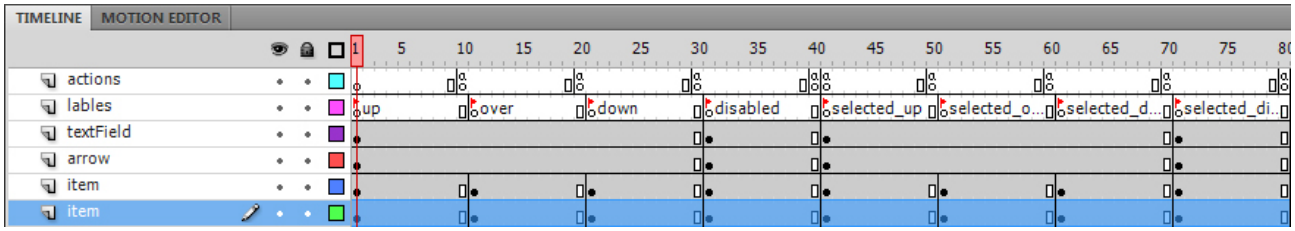
- ***textField*** ( ) TextField
- ***focusIndicator***: ( ) MovieClip
  “show” “hide”

2.4.6.3

DropdownMenu ToggleButton CheckBox

- ***up***

- *over*
- *down*
- *disabled*
- *selected_up*
- *selected_over*
- *selected_ down*
- *selected_disabled*

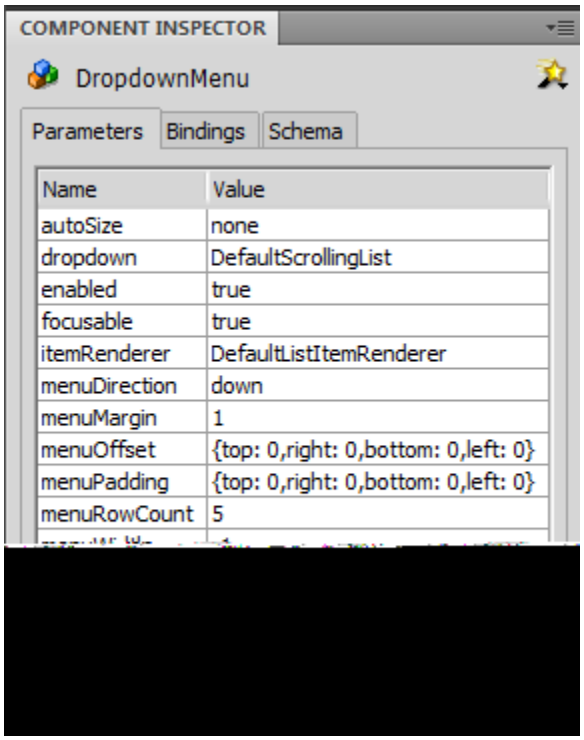


44: DropdownMenu

DropdownMenu Button

DropdownMenu CLIK

2.4.6.4



DropdownMenu

autoSize	DropdownMenu autoSize autoSize="none"
Dropdown	ScrollingList TileList DropdownMenu
Enabled	Disables the component if set to false.
Focusable	false
menuDirection	"up" "down"
menuMargin	
menuOffset	
menuPadding	
menuRowCount	
menuWidth	If set, this number will be enforced as the width of the drop down's list.
thumbOffset	
scrollbar	
Visible	false

2.4.6.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

DropdownMenu *change* Button

ComponentEvent.SHOW	visible	true

ComponentEvent.HIDE	visible	false
ComponentEvent.STATE_CHANGE		
Event.SELECT		
FocusHandlerEvent.FOCUS_IN		
FocusHandlerEvent.FOCUS_OUT		
ListEvent.INDEX_CHANGE		

- *itemRenderer* Ia-4(aAI)-4(a-4sa-4(t(a-4(t(r)-5men)-R

	controllerIdx uint *isKeyboard* true - (Keyboard)/ (Gamepad) false – *isRepeat* true – autoRepeating false –
ButtonEvent.DRAG_OVER	DropdownMenu *controllerIdx* uint
ButtonEvent.DRAG_OUT	DropdownMenu *controllerIdx* uint
ButtonEvent.RELEASE_OUTSIDE	*controllerIdx* uint

2.5

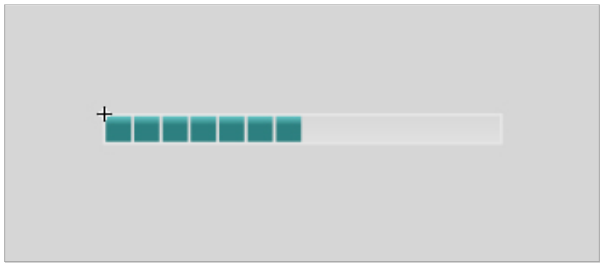
The Scaleform CLIK

StatusIndicator ProgressBar StatusIndicator

ProgressBar

45: ***Mercenaries 2*** **Faction**

2.5.1 StatusIndicator



46: **StatusIndicator.**

StatusIndicator component (scaleform.clik.controls.StatusIndicator)

2.5.1.1

StatusIndicator

2.5.1.2

CLIK StatusIndicator MovieClip

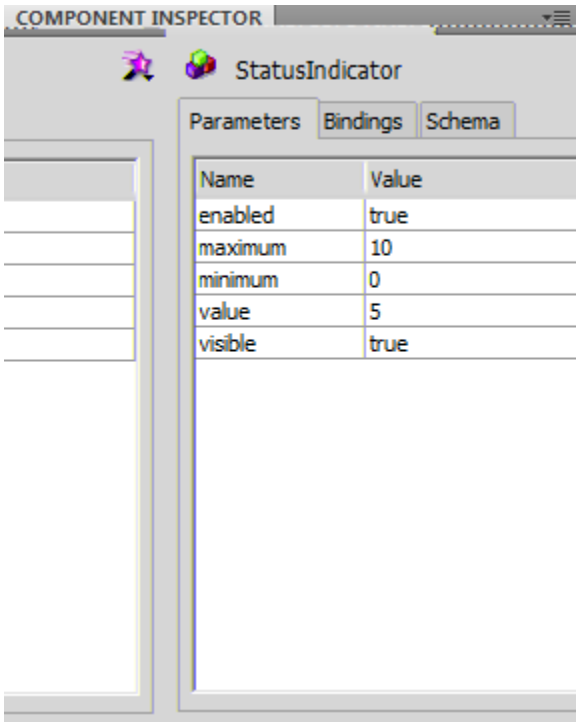
stop()

gotoAndStop()

2.5.1.3

StatusIndicator

2.5.1.4



StatusIndicator MovieClip

Visible	false
Enabled	false
Value	
Minimum	
Maximum	

2.5.1.5

Event Event

- ***type***
- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

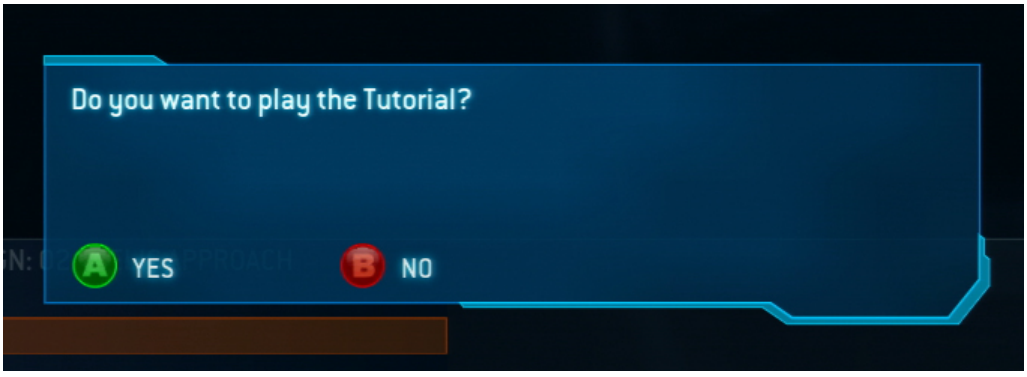
StatusIndicator

ComponentEvent.SHOW	visible	true
ComponentEvent.HIDE	visible	false

2.6

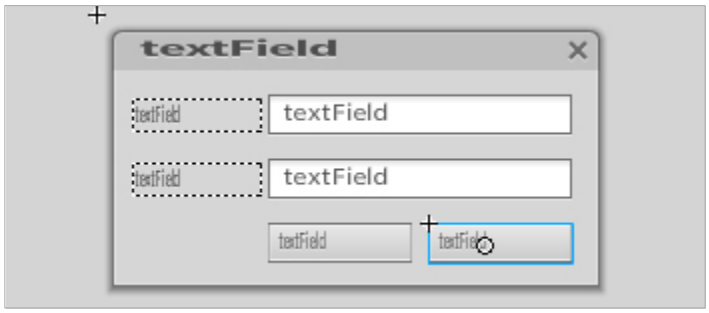
Scaleform CLIK UI Window

DragSlot CLIK CLIK UI CLIK CLIK DragManager scaleform.clik.managers.DragManager.as



48: *Halo Wars* Window

2.6.1 Window



49: Window

CLIK Window (scaleform.clik.controls.Window) Alert

PopUpManager PopUpManager.showModal()

2.6.1.1

Dialog

2.6.1.2

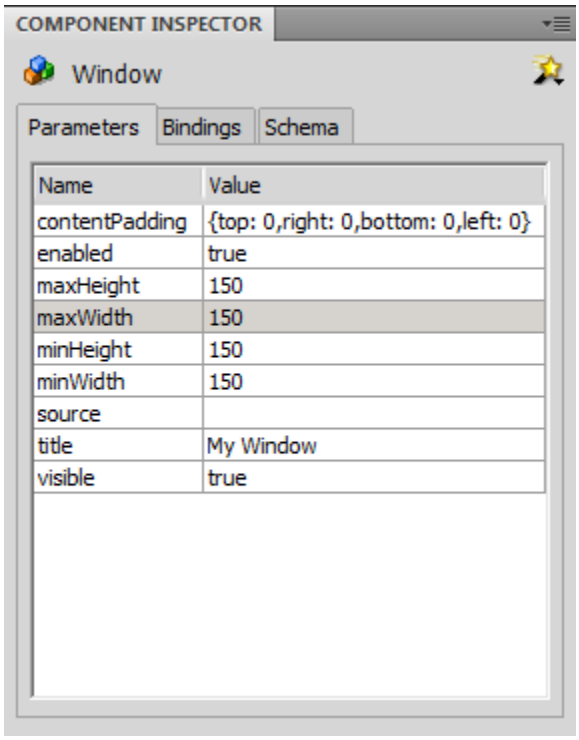
CLIK Dialog MovieClip :

- ***closeBtn*** CLIK Button
- ***titleBtn*** CLIK Button
- ***okBtn*** CLIK Button “Accept” “OK”
- ***resizeBtn***: CLIK Button
- ***background*** MovieClip
- ***hit*** MovieClip hitArea

2.6.1.3

Dialog MovieClip

2.6.1.4



Window MovieClip

contentPadding	
Enabled	false
maxHeight	
maxWidth	
minHeight	
minWidth	
Source	
Title	titleBtn
Visible	false

2.6.1.5

Event Event

- ***type***

- ***target***
- ***currentTarget*** Event
- ***eventPhase*** EventPhase.CAPTURING_PHASE EventPhase.AT_TARGET EventPhase.BUBBLING_PHASE
- ***bubbles***
- ***cancelable***

Window

ComponentEvent.SHOW	visible	true
ComponentEvent.HIDE	visible	false

```
myWindow.addEventListener(ComponentEvent.HIDE, onWindowClosed);
function onWindowClosed(e:ComponentEvent) {
      // Do something.
}
```

3

Scaleform CLIK

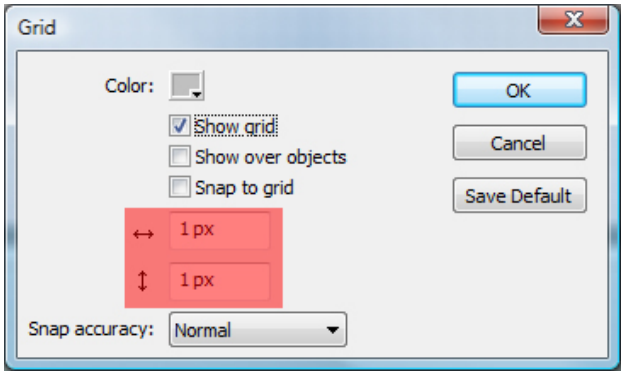
3.1

Scaleform CLIK

3.1.1

CLIK Flash

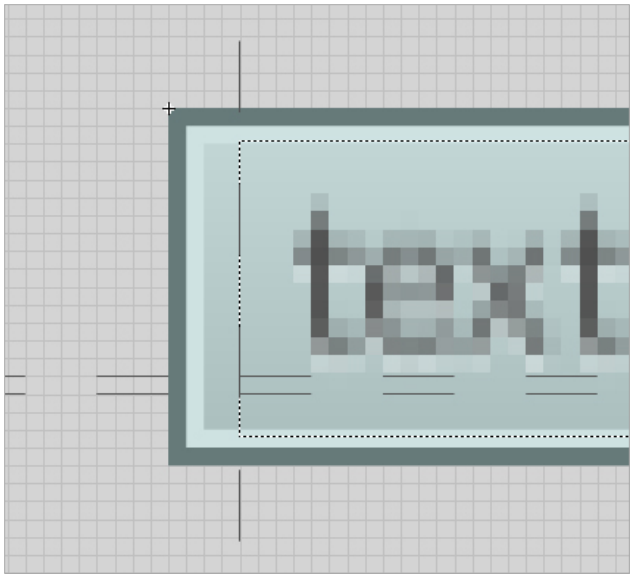
1. Flash
2.
3. Flash
4.
5. 1px
6. OK



51: Grid

SWF 2

2



52:

3.1.2

Flash

Scaleform

Flash Photoshop® PNG

3.1.3

–

3.1.4

Flash

3.1.5

3.1.6 2

2 2

- 16x16
- 32x32
- 64x64
- 128x128
- 128x256
- 256x256
- 512x256
- 512x512

3.2

Flash Scaleform CLIK

3.2.1

Flash

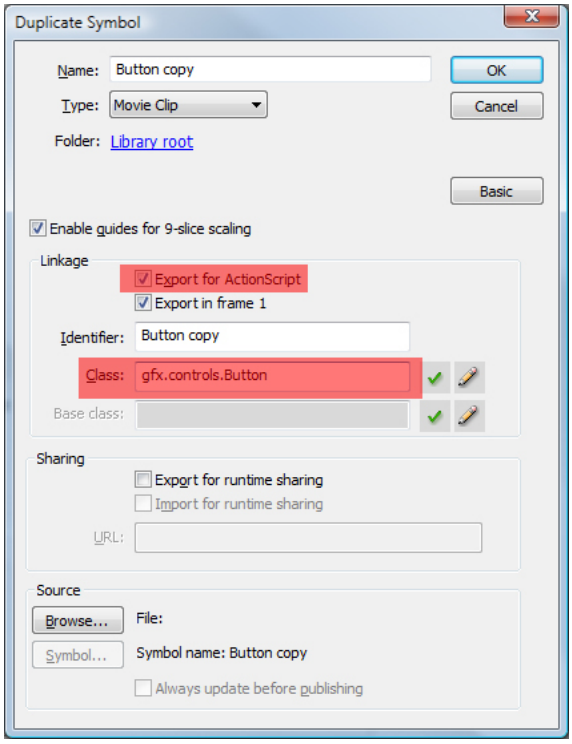
3.2.1.1 1

FLA CLIK FLA

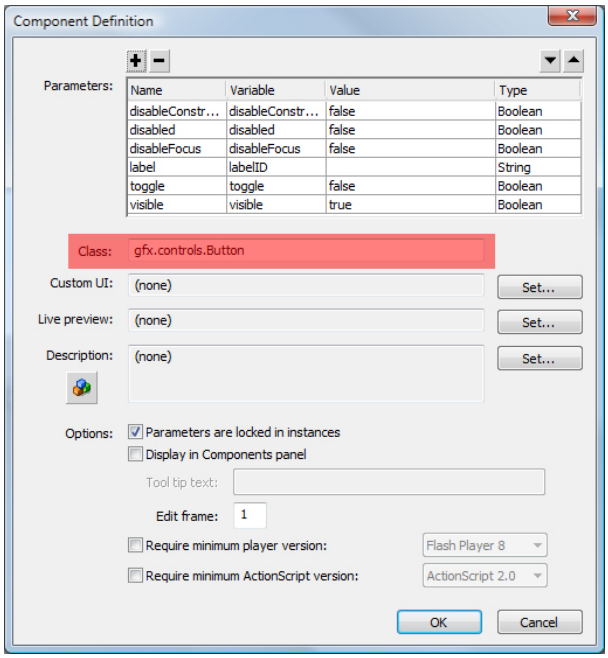
1. *CLIK_Components.fla*
2. FLA *Copy*
   FLA *Paste*
3. FLA *Rename* ‘Button’
4. FLA *Properties*
5. *Identifier* 3
6. *Paste*

3.2.1.2 *2*

1. *library* *Properties*
2. *Properties* scaleform.clik.controlsButton *Class* (CTRL+C)
3. *Cancel*
4. *Duplicate*
5. *Duplicate Symbol* *Export* *ActionScript*
6. *Class* (CTRL+V) textField
7. OK
8. *library* *Component Definition*
9. *Class* textField (CTRL+V) textField
10. OK



53:



54:

3.3

Scaleform CLIK

CLIK

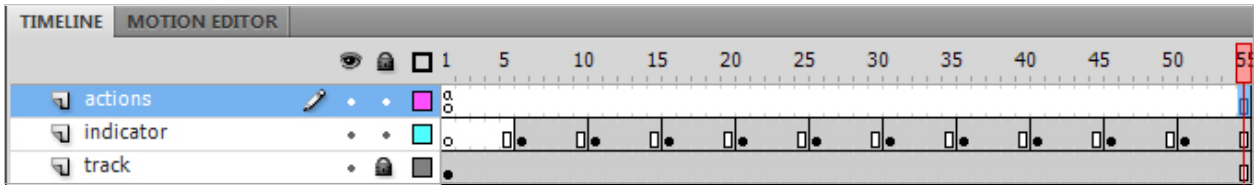
CLIK Flash FLA

- Flash
- Flash
- Photoshop Illustrator® Flash

CLIK

1. Photoshop StatusIndicator

2. PSD
3. Flash *File* *Import -> Import to Stage*
4. StatusIndicator PSD
5. *Import* *Convert layers to*: 'Layers'
6. OK
7. Flash PSD 10
PSD 10 1 10
1 10 *layer 1*
8.
9.
10. *layer 1* 6
11. *Insert* *layer 1* 11
12. *layer 2* (Ctrl+X)
13. *layer 1* 11 (Ctrl+Shift+V) *layer 1*
14. 10 (*layer 1*)
5 *layer 2* 11
layer 3 *16* *layer 4* 21
10 PSD 51
15. 2-10
16. *Remove Frames*
55
17. *indicator layer*
18. *track* *track*
19. *layer 1* 'indicator'
20. *indicator* *track* *actions layer* *track* *indicator*



58:

21. StatusIndicator
22. 1 10

23.
24.

59: StatusIndicator

3.4

Scaleform CLIK

3.4.1

Scaleform SWF Scaleform
Flash UI

3.4.2

Flash Scaleform Scaleform
Scaleform Scaleform
textFidId Slate Mobile
Slate Mobile CJK ASCII
Slate Mobile CJK

textFields Scaleform 3.4.4
Scaleform Scaleform
UI

3.4.3 textField

textFields
textFidId (Window > Property

Inspector) *Embed* *OK*
FLA FLA

3.4.4

Scaleform
SWF SWF Scaleform

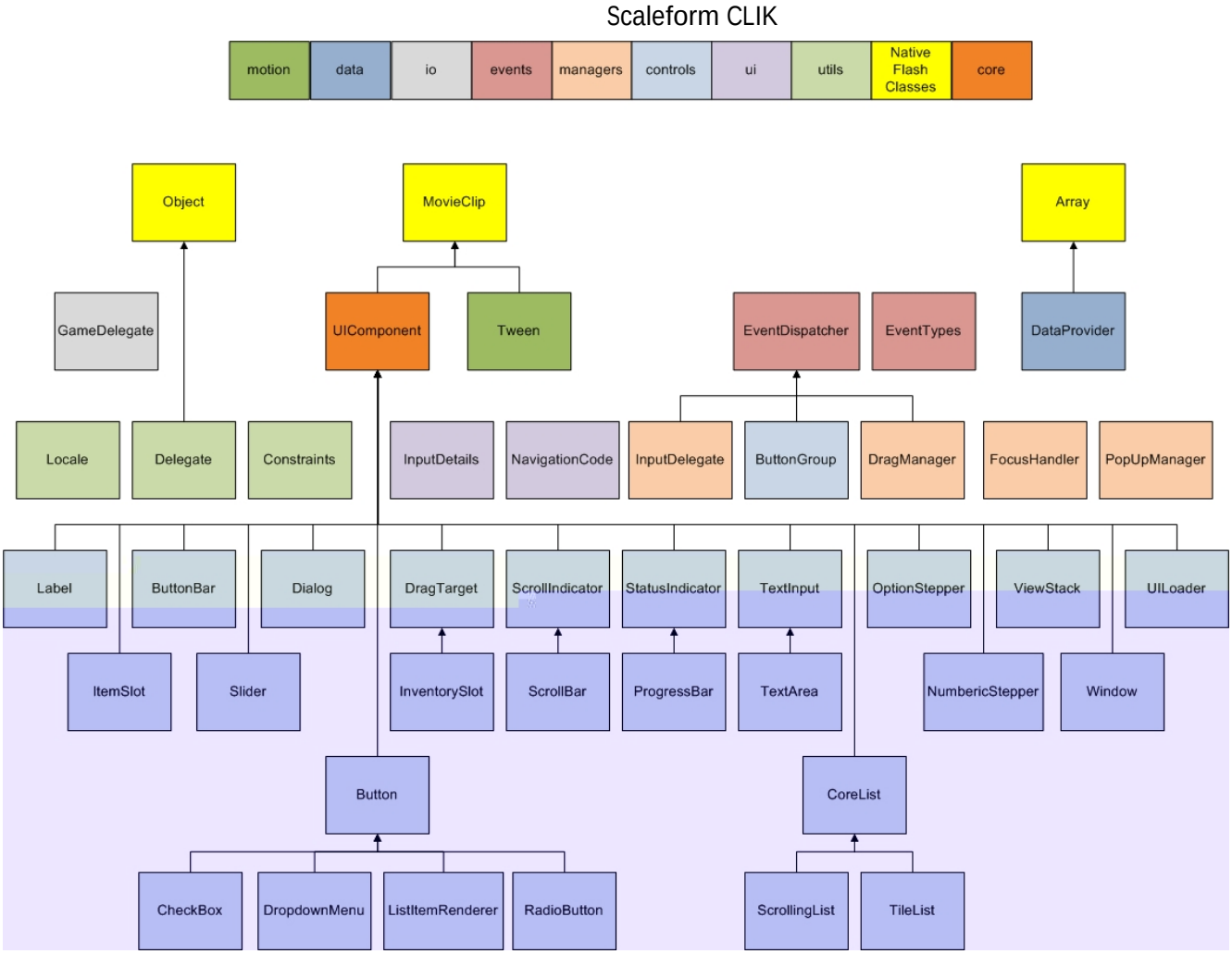
Scaleform textField
gfxfontlib.fla swf
Scaleform

Scaleform
fontconfig.txt

fontconfig.txt Scaleform
‘$TITLE’
$TITLE=Scaleform ‘Scaleform’

Scaleform Scaleform

4



60: Scaleform CLIK

4.1 UI UIComponent

Scaleform CLIK UIComponent (scaleform.clik.core.UIComponent). CLIK Scaleform UIComponent

UIComponent Flash 8 MovieClip Flash MovieClip / UIComponent

- ***enabled;***
- ***visible;***
- ***focused;***

- ***focusable;***
- ***width;***
- ***height;***
- ***scaleX;***
- ***scaleY;***
- ***displayFocus***: true

  focusTarget: Settings this property to another MovieClip will cause that MovieClip to receive focus rather than this component if focus is ever set to this MovieClip. For more information see the Focus Handling section.

UIComponent

- ***configUI***:
- ***draw***:
- ***changeFocus***:
- ***scrollWheel***:

UIComponent EventDispatcher

4.1.1

onLoad()

- MovieClip
- configUI()
- validateNow() draw()

4.2

Scaleform CLIK

4.2.1

CLIK Button ListItemRenderer RadioButton CheckBox ScrollBar Button CLIK Button (scaleform.clik.controls.Button) Button

- ***up***
- ***over***

- ***down***
- ***disabled***

Button CLIK Button “selected_”

- ***up***
- ***over***
- ***down***
- ***disabled***
- ***selected_up***
- ***selected_over***
- ***selected_ down***
- ***selected_disabled***

CLIK CLIK

CLIK getStatePrefixes()

```
protected function getStatePrefixes():Vector.<String> {
      return _selected ? statesSelected : statesDefault;
}
```

CLIK “selected_” getStatePrefixes()

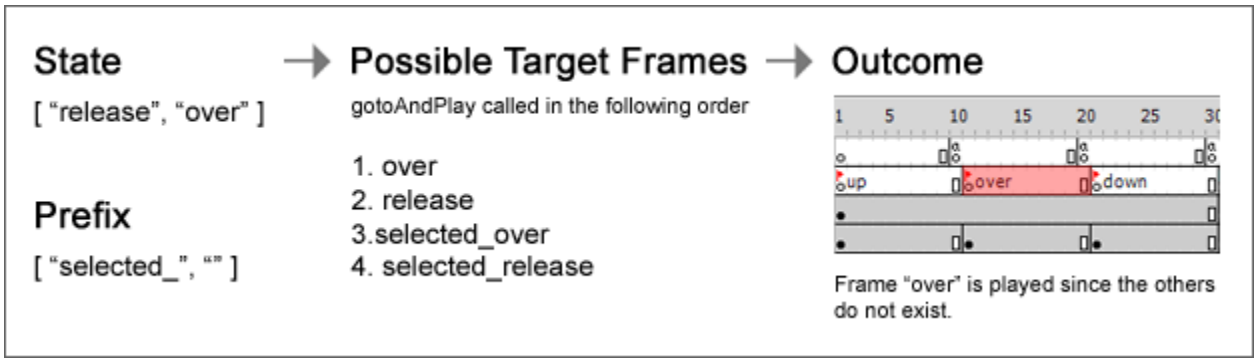
stateMap

CLIK

```
protected var _stateMap:Object = {
            up:["up"],
            over:["over"],
            down:["down"],
            release: ["release", "over"],
            out:["out","up"],
            disabled:["disabled"],
            selecting: ["selecting", "over"],
```

```
                toggle: ["toggle", "up"],
                kb_selecting: ["kb_selecting", "up"],
                kb_release: ["kb_release", "out", "up"],
                kb_down: ["kb_down", "down"]
            }
```

getStatePrefixes()



61:

4.2.2

CLIK

ScrollingList OptionStepper Slider TextArea

- ***default***
- ***focused***
- ***disabled***

4.2.3

- ***default***
- ***disabled***

4.2.4

StatusIndicator ProgressBar

50

0 10 5 50% gotoAndStop() 25 50 50% updateValue()

“ ” “ ” “ ” “ ” “ ”

4.3

Scaleform CLIK *event model* “ ”
ActionScript 3
CLIK

ActionScript 3
http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/Event.html

4.3.1

4.3.1.1

addEventListener()
removeEventListener()

CLIK
Event CLIK scaleform.clik.events

addEventListener() **useCapture** **priority** **useWeakReference**

addEventListener()

```
buttonInstance.addEventListener(ButtonEvent.CLICK, onButtonClick, false,
0, true);
```

addEventListener()
http://help.adobe.com/en_US/FlashPlatform/reference/actionscript/3/flash/events/EventDispatcher.html#addEventListener()

```
buttonInstance.addEventListener(ButtonEvent.CLICK, onButtonClick, false,
0, true);
```

```
      function onButtonClick (e:Event):void {
            buttonInstance.removeEventListener(ButtonEvent.CLICK,
onButtonClick, false);
      }
```

buttonInstance ButtonEvent.CLICK
onButtonClick

Event CLIK

4.3.1.2

CLIK dispatchEvent()

```
      dispatchEvent(new Event(Event.CHANGE));
```

4.4

ActionScript 3

```
import flash.utils.getDefinitionByName;
import scaleform.clik.controls.Button;

public var myButton:Button;

//       Class
var classRef:Class = getDefinitionByName("ButtonLinkageID") as Class;
//
if (classRef != null) { myButton = new classRef () as Button; }
myButton.addEventListener(ButtonEvent.CLICK, onButtonClick, false, 0, true);
```

.FLA Class “ButtonLinkageID” Base Class
scaleform.clik.controls.Button

.FLA

4.5

Scaleform CLIK

CLIK FocusHandler

(scaleform.clik.managers.FocusHandler) FocusHandler

FocusHandler

Scaleform

ActionScript ActionScript

- myUIComponent.focused = 1; // 1 1
- stage.focus = myMovieClip; // MovieClip Stage
- FocusHandler.getInstance().setFocus(myMovieClip); // FocusHandler

4.5.1

InteractiveDisplayObject

Tab true

tabEnabled ***mouseEnabled*** *false* MovieClips Scaleform Flash

CLIK tabEnabled

mouseEnabled

tabEnabled mouseEnabled

UIComponent.***focusable*** AS3 MovieClip.focusEnabled

false UIComponent CLIK ***focusable***

tabEnabled mouseEnabled tabEnabled

mouseEnabled

ScrollBar

FocusHandler

displayFocus

true Slider

UIComponent.changeFocus()

```
function changeFocus():void {
      track.displayFocus = _focused;
}
```

tab

tabEnabled false

```
background.tabEnabled = false;
```

4.5.2

ScrollingList OptionStepper ButtonBar

Flash Scaleform Selection

1. tabEnabled = mouseEnabled = focusable = false
   OptionStepper
2. focusable= true
3. displayFocus true changeFocus

4.6

Scaleform CLIK Flash 8 MovieClip MovieClip

CLIK

4.6.1

InputDelegate class (scaleform.clik.managers.InputDelegate)

InputDelegate InputDetails (scaleform.clik.ui.InputDetails)

InputDelegate.readInput()

InputDetails InputDelegate

InputDetails

- “key”
-
-

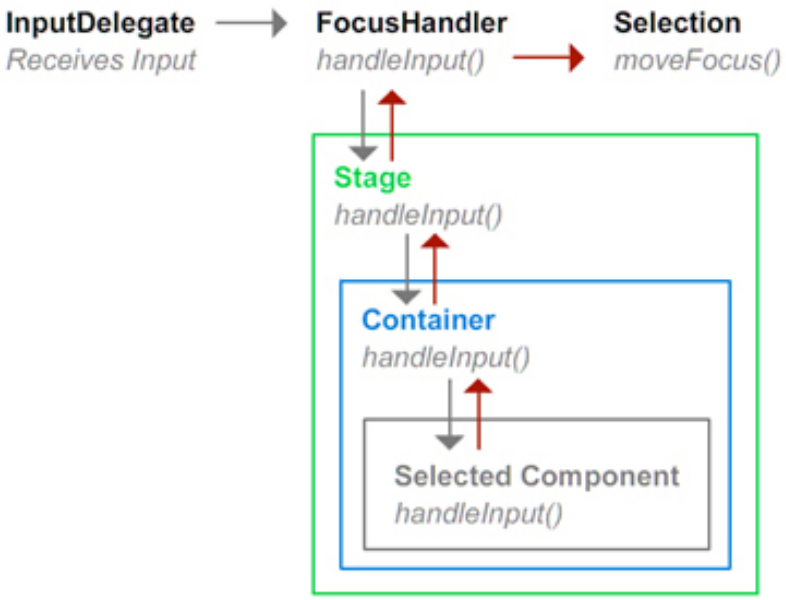
InputValue.KEY_UP (“keyUp”)

InputValue.KEY_DOWN (“keyDown”)

- navEquivalent(navigation equivalent) NavigationCode.UP (“up”), NavigationCode.DOWN (“down”), NavigationCode.LEFT (“left”), or NavigationCode.RIGHT (“right”), “right” NavigationCode (scaleform.clik.ui.NavigationCode)

FocusHandler InputDelegate InputEvent
InputEvent bubble

the handleInput()



62: handleInput()

InputEvent ***handled*** true
event.isDefaultPrevented()

```
override public function handleInput(event:InputEvent):void {
      // Check whether the event has already been handled by another component.
      if (event.isDefaultPrevented()) { return; }
      var details:InputDetails = event.details;
```

```
      switch (details.navEquivalent) {
          case NavigationCode.ENTER:
              if (details.value == InputValue.KEY_DOWN) {
                    // Do something.
                    event.handled = true;
              }
              break;
          default:
              break;
      }
}
```

InputEvent ActionScript 3 (bubble) .bubble
ScrollingList InputEvent
Enter ListItemRenderer ButtonEvent.CLICK

```
override public function handleInput(event:InputEvent):void {
      if (event.isDefaultPrevented()) { return; }
      // Pass on to selected renderer first
      var renderer:IListItemRenderer = getRendererAt(_selectedIndex,
_scrollPosition);
      if (renderer != null) {
            // Since we are just passing on the event, it won't bubble, and should
properly stopPropagation.
            renderer.handleInput(event);
            if (event.handled) { return; }
      }
      . . .
}
```

InputDelegate.instance
FocusHandler

```
      InputDelegate.instance.addEventListener(InputEvent.INPUT, handleInput);
      function handleInput(e:InputEvent):void {
            var details:InputDetails = e.details;
            if (details.value = Key.TAB) { doSomething(); }
      }
```

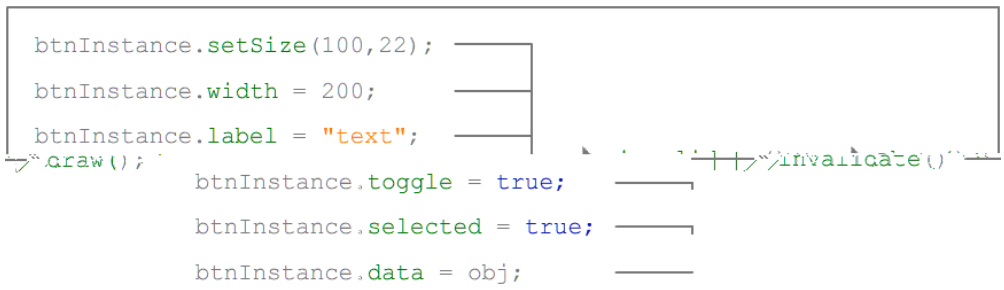
InputDelegate InputDelegate
W A S D

4.6.2

Nintendo Wii™ CLIK
SWF

```
myButton.addEventListener(ButtonEvent.CLICK, onCursorClick);
function onCursorClick(event:ButtonEvent):void {
	trace(event.controllerIdx);
}
```

4.7



63: CLIK

4.7.1

UIComponent.invalidate()

UIComponent.draw()

draw() isInvalid()

invalidate()

draw()

isInvalid() draw() invalidate()

invalidate() invalidateData(), invalidateSize()

invalidate() (Event.RENDER) (Event.ENTER_FRAME) draw()

draw()

UIComponent.validateNow()

draw()

4.8

Scaleform CLIK

1.

2.

scale9grid

Flash scale9Grid “ ”
Flash/Flex container

CLIK Scaleform scale9Grid
CLIK

container

Flash IDE setSize()
Flash IDE
MovieClips LivePreviews Scaleform Player
CLIK

4.8.1 Scale9Grid

Scaleform Player Scale9Grid MovieClip
scale9grid Flash MovieClips scale9grid Flash
Player Scaleform

MovieClip scale9grid
Scale9grid

4.8.2

Constraints (scaleform.clik.utils.Constraints)
ScrollBar
Constraints

ScrollBar configUI()
draw()
draw()

```
override protected function initialize():void {
      super.initialize();

      // The upArrow doesn't need a constraints, since it sticks to top left.
      if (downArrow) {
            constraints.addElement("downArrow", downArrow, Constraints.BOTTOM);
      }
      constraints.addElement("track", track, Constraints.TOP |
                                             Constraints.BOTTOM);
}

override protected function preInitialize():void {
     constraints = new Constraints(this, ConstrainMode.REFLOW);
}

protected function draw():void {
     constraints.update(_width, _height);
}
```

4.9

dataProvider dataProvider
Scaleform CLIK IDataProvider (scaleform.clik.interfaces.IDataProvider)
API

dataProvider

dataProvider CoreList (ScrollingList, TileList) OptionStepper
DropdownMenu

4.9.1

DataProvider (scaleform.clik.data.DataProvider) initialize()
dataProvider ActionScript dataProvider
dataProvider
dataProvider IDataProvider

```
myComponent.dataProvider = new DataProvider([   "data1",
                                                4.3,
                                                {label:"anObjectElement", value:6}]);
```

dataProvider

- *length*:
- *requestItemAt*: dataProvider
  OptionStepper
- *requestItemRange:* dataProvider
  ScrollingList
- *indexOf:*
- *invalidate:* dataProvider
  “dataChange” dataProvider

dataProvider ActionScript 2

dataProvider ExternalInterface.call

4.10

Scaleform CLIK Tween (scaleform.clik.motion.Tween) Tween
Scaleform CLIK Tween
fl.transitions.easing.*
tween Tween() Sprite
tween tween Tween
/

- *ease* Function
- *easeParam* Object
- *onComplete* Function Tween (closure)
- *onChange* Function Tween / (tick/frame)
- *data* Object Tween Tween

- *nextTween* Tween Tween Tween
  Tween pause=false
- *frameBased* Boolean
- *delay* Number tween x
- *fastTransform* Boolean

- *paused* Boolean Tween

Tween MovieClip
Tweens

4.10.1

Tween MovieClip MovieClip tween

tween onComplete
Tween Tween
Tween

```
import fl.transitions.easing.*;
import scaleform.clik.motion.Tween;

// Perform a 1 second tween from the current horizontal position and
// alpha values to the ones specified
var myTween:Tween = new Tween(300,
                              myMovieClip,
                              { x:100, y:100, alpha:0 },
                              { ease:Strong.easeOut,
                              onComplete:onCompleteCallback} );

function onCompleteCallback(t:Tween):void { // Do something. }
```

4.11

CLIK
CLIK UI CLIK

Scaleform v4.0.14 CLIK AS3 API
CLIK AS3 Scaleform

CLIK

- **scaleform.clik.layout.Layout** Sprite Sprite
  .layoutData

- **scaleform.clik.layout.LayoutData** Sprite

CLIK

1. Sprite/MovieClip Stage DisplayObjectContainer
2. LayoutData Sprite/MovieClip .layoutData
3. Sprite/MovieClip LayoutData
4. Layout 1 Sprites/MovieClips DisplayObjectContainer

Layout Layout Sprite MovieClip Sprite MovieClip x y .layoutData

Scaleform\Resources\AS3\CLIK\demos SDK Layout_Main CLIK Scaleform SDK Layout_Main MovieClip Layout_FullScreen_Demo FxPlayer Stage Layout_FullScreen_Demo Layout

4.11.1

(Layout) / Sprite

myWindow.myMovieClip myWindow.layoutInstance layoutInstance myMovieClip

Sprite Sprite LayoutData .layoutData .identifier Sprite LayoutData .layoutIdentifier

8-1(e( )Tji-8(a)-5(yo)-12(u)-7(t)-7(In)-13(st)-7(a)-5D1(8-1(e(_0 1 Tf3.978 0 Td( )Tj-0Tm34 58318B0 1 Tf11C2_10.27860y)7

Sprite LayoutData Sprite /
Sprite Layout

public function resortManagedSprites():void
layoutData.layoutIndex Sprite
layoutIndex

Layout Flash (Symbol)
Stage Layout Symbol 0
Layout Layout .rect
Layout

Layout Sprite/MovieClip
LayoutData

4.11.1.1

Rect	" " x y Layout != 0 Symbol Stage Layout MovieClip x y Layout == 0 addChild() Symbol Layout x y .rect Rectangle
tiedToStageSize	true – Layout Stage false
tiedToParent	true – Layout false
Hidden	true - Layout false Layout visible = false

4.11.2 LayoutData

Sprite scaleform.clik.layout.Layout

4.11.2.1

alignH	Sprite LayoutMode.ALIGN_NONE LayoutMode.ALIGN_LEFT LayoutMode.ALIGN_RIGHT
alignV	Sprite LayoutMode.ALIGN_NONE LayoutMode.TOP LayoutMode.BOTTOM.property TextFieldAutoSize.NONE
offsetH	Layout relativeToH Sprite (-1) Layout Stage Layout/Sprite

offsetV	Layout relativeToH Sprite (-1) Layout Stage Layout/Sprite
offsetHashH	(Hash Table) Layout Stage (Layout.tiedToStageSize == true) Layout Stage 4:3 "LayoutData.offsetHashH[LayoutData.ASPECT_RATIO_4_3] = 70;" Sprite 70px
offsetHashV	(Hash Table) Layout Stage (Layout.tiedToStageSize == true) Layout Stage 4:3 "LayoutData.offsetHashV[LayoutData.ASPECT_RATIO_4_3] = 70;" Sprite 70px
relativeToH	Sprite Sprite null Sprite Layout
relativeToV	Sprite Sprite null Sprite Layout
layoutIndex	Layout Sprite Sprite – relativeToH relativeToV Sprite Sprite Layout layoutIndex (-1)
layoutIdentifier	LayoutData Layout DisplayObjectContainer Layout (null) Layout Layout

4.12

Scaleform CLIK PopUpManager (scaleform.clik.managers.PopUpManager)

4.12.1

PopUpManager createPopUp() MovieClip CLIK ID ID initProperties PopUp x y relativeTo Sprite Sprite

```
import scaleform.clik.managers.PopUpManager;
PopUpManager.createPopUp("PopupLinkageID", {alpha:.5}, 100, 100
myMovieClip);
```

Scaleform InteractiveObjectEx.setTopmostLevel()
root SWF

topmostLevel

InteractiveObjectEx.setTopmostLevel() Z
topmostLevel
true

4.13

DragManager (scaleform.clik.managers.DragManager)

DragEvent.DRAG_STATE
DragManager.handleStartDragEvent()
DragManager MouseEvent.MOUSE_UP
DragManager.handleEndDragEvent()

4.13.1

64: DragDemo

DragManager IDragSlot UI IDragSlot
Scaleform CLIK DragTarget (scaleform.clik.controls.DragTarget)
DragTarget UIComponent CLIK DragManager

Scaleform CLIK IDragSlot DragSlot (scaleform.clik.controls.DragSlot)
DragSlot DragManager dragBegin dragEnd
DragTarget DragSlot
MovieClip/Sprite/Bitmap DragSlot DragSlot

CLIK DragSlot DragManager DragSlot
InventorySlot ItemSlot CLIK

DragSlot IDragSlot
DragSlot MMO UI
DragSlot UI

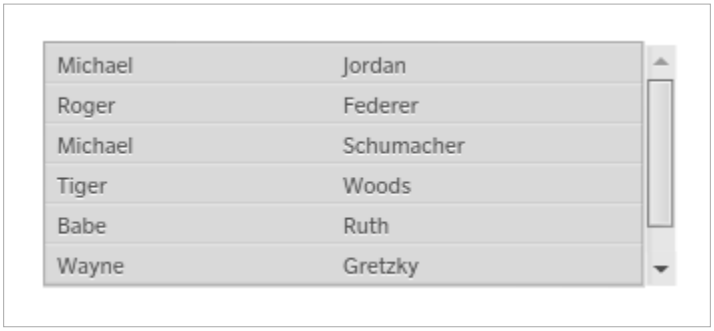
5

Scaleform CLIK

5.1

Scaleform CLIK

5.1.1 textField ListItem Renderer



65: ScrollingList

ListItemRenderer ListItemRenderer
textField textField

ListItemRenderer ListItemRenderer
ScrollingList textFields
dataProvider

```
list.dataProvider = [{fname: "Michael", lname: "Jordan"},
                     {fname: "Roger",   lname: "Federer"},
                     {fname: "Michael", lname: "Schumacher"},
                     {fname: "Tiger",   lname: "Woods"},
                     {fname: "Babe",    lname: "Ruth"},
                     {fname: "Wayne",   lname: "Gretzky"},
                     {fname: "Usain",   lname: "Bolt"}];
```

fname lname textField

ListItemRenderer textField textField
ListItemRenderer MyItemRenderer

ListItemRenderer *MyItemRenderer.as*

FLA

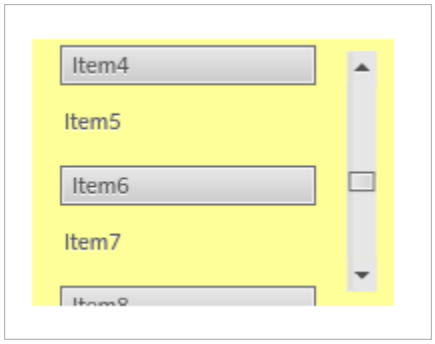
```
import scaleform.clik.controls.ListItemRenderer;

class MyItemRenderer extends ListItemRenderer {

	public var textField1:TextField;
	public var textField2:TextField;

	public function MyItemRenderer() { super(); }

	override public function setData(data:Object):void {
		this.data = data;
		textField1.text = data ? data.fname : "";
		textField2.text = data ? data.lname : "";
	}

	override protected function updateAfterStateChange():void {
		super.updateAfterStateChange();
		textField1.text = data ? data.fname : "";
		textField2.text = data ? data.lname : "";
	}
}
```

ListItemRenderer setData updateAfterStateChange setData (ScrollingList TileList) ListItemRenderer textField MyItemRenderer textField updateAfterStateChange ListItemRenderer textField

ListItemRenderer dataProvider ListItemRenderer ListItemRenderer textFild ‘textField1’ ‘textField2’ MyItemRenderer MyItemRenderer itemRenderer ‘ListItemRenderer’ ‘MyItemRenderer’

FLA dataProvider fname lname

5.1.2



66: **CLIK**

CLIK ScrollBar

- ***actions***: ActionScript
- ***scrollbar***: CLIK ScrollBar ‘sb’
- ***mask*** MovieClip
- ***content***
- ***background***

‘content’ y 0

ActionScript

code

```
// 133 is the view size (height of the mask)
sb.setScrollProperties(1, 0, content.height - 133);
sb.position = 0;

sb.addEventListener(Event.SCROLL, onScroll, false, 0, true);
function onScroll(e:Event) {
      content.y = -sb.position;
}
```

setScrollProperties

6

1. Flash Studio

```
1152: A conflict exists with inherited definition scaleform.clik... in
namespace public.          public                    scaleform.clik...
```

.FLA

"Automatically Declare Stage Instances" ActionScript .FLA .FLA

CLIK CLIK Stage
ActionScript

File -> Publish Settings -> "Flash" tab -> Actionscript 3.0 Settings
“Automatically declare stage instances” .FLA "Automatically Declare Stage Instances"
Flash Studio .FLA

2. .FLA CLIK Stage (inspectable)

```
1046: Type was not found or was not a compile-time constant:
                    ...
```

-
- ActionScript
- ActionScript –
  ActionScript myButton:Button (scaleform.clik.controls.)
  myButton:MyButtonSymbol MyButtonSymbol myButton

3. .SWF

```
Symbol ‘MySymbol’, Layer ‘actions’, Frame 10, Line 1 -- 1024: Overriding a function
that is not marked for override.
```

10. DataProvider now requires that you provide the target array to DataProvider's constructor (myComponent.dataProvider = new DataProvider(myArray);), rather than simply setting myComponent.dataProvider = myArray;
11. ButtonBar will now only create Buttons that fit within its own size. If your ButtonBar's size is 200px wide and each Button is 150px wide, only one Button will be displayed. This allows users to resize their ButtonBar and have the Buttons within be added and removed dynamically.
12. UILoader has been removed. This functionality is now handled by the Flash MovieClip/Sprite and Loader classes.
13. UIComponent.SoundMap, introduced in CLIK 3.3, has been removed for CLIK AS3.